Paleomagnetism and the evolution of fluids in the Proterozoic Athabasca Basin, northern Saskatchewan, Canada[1]

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In the Athabasca Basin, diagenetic hematite of variable paragenesis occurs throughout the sandstones and underlying paleoregolith. This hematite carries three distinct, single-component magnetizations: A ($D = 158°$, $I = 62°$, $\alpha_{95} = 5°$, $n = 21$); B ($D = 11°$, $I = -36°$, $\alpha_{95} = 7°$, $n = 6$); and C ($D = 18°$, $I = 79°$, $\alpha_{95} = 3°$, $n = 27$). In some areas of the sandstones, such as near reactivated fault zones, the diagenetic hematite has been altered to goethite which yields a very low-intensity, incoherent D magnetization. Ages for the A, B, and C magnetizations, inferred from comparisons with paleomagnetic directions in Precambrian rocks whose ages are known approximately, are 1750–1600, 1600–1450, and about 900 Ma, respectively. The A magnetization is carried by the earliest formed hematite, and its estimated age compares well with U–Pb ages of 1650–1700 Ma for early diagenetic apatite. U–Pb and Rb–Sr ages of approximately 1500 and 900 Ma for uraninite and illite coeval with hematite that carries the B and C magnetizations compare well with their ages estimated from paleomagnetism. The development of B magnetization appears to be coeval with high-grade, unconformity-type uranium mineralization.

Petrographic and field relationships indicate that the A magnetization is carried by hematite formed during initial diagenesis of the Athabasca sandstones, the B magnetization is carried by hematite formed during peak diagenesis, and the C magnetization is carried by hematite formed during subsequent high-temperature hydrothermal alteration. The incoherent D magnetizations have resulted from degradation of hematite to goethite as a result of incursion of low-temperature meteoric waters along fault zones that have been continuously reactivated since the late Precambrian. $\delta^{18}O$ values of clay minerals and of the coeval hematite which carries the B and C magnetization indicate that they were formed from a fluid having temperatures of 150–200°C and $\delta^{18}O$ values near 1.0‰. Fluids that deposited the early formed hematite carrying the A magnetism are relatively ^{18}O depleted, with values of approximately 0.8‰ and somewhat lower temperatures of 120–160°C. Intermingling of A, B, and C magnetizations indicates either that hematite may be deposited by one fluid and reprecipitated by a subsequent fluid, or that fluid flow was controlled by local variations in permeability. Evidently, fluid flow has been episodic and basin wide and has occurred over a time span on the order of 10^8 years. It is suggested that the stratigraphy of the sandstones controlled the basin-wide lateral migration of the basinal fluids and that faults facilitated interformational fluid flow.

Dans le bassin d'Athabasca, l'hématite diagénétique de paragenèse variable est distribuée au travers les grès et le paléorégolithe sous-jacent. Cette hématite porte trois composantes discrètes d'aimantation bien distinctes : A ($D = 158°$, $I = 62°$, $\alpha_{95} = 5°$, $n = 21$); B ($D = 11°$, $I = -36°$, $\alpha_{95} = 7°$, $n = 6$); et C ($D = 18°$, $I = 79°$, $\alpha_{95} = 3°$, $n = 27$). Dans certaines parties de ces grès, comme dans les zones de failles réactivées, l'hématite diagénétique altérée en goethite fournit une aimantation D incohérente et de très faible intensité. Les âges des aimantations A, B et C, déduites de comparaisons faites avec les directions paléomagnétiques dans les roches précambriennes dont les âges sont connus approximativement, sont de 1750–1600 Ma pour A, 1600–1450 Ma pour B et environ 900 Ma pour C. L'aimantation A est portée par l'hématite qui a précipité en premier, et son âge estimé se compare raisonnablement avec les âges U–Pb de 1650–1700 Ma fournis par l'apatite diagénétique précoce. Les âges U–Pb et Rb–Sr pour l'uraninite et l'illite, contemporaines de l'hématite qui porte les aimantations B et C, d'environ 1500 et 900 Ma, correspondent bien aux âges estimés par paléomagnétisme. L'acquisition de l'aimantation B semble avoir été contemporaine de la minéralisation élevée en uranium et de type discordante.

Les relations de terrain et l'étude pétrographique révèlent que l'aimantation A est portée par la magnétite formée durant le stade de la diagenèse précoce des grès d'Athabasca, que l'aimantation B est portée par l'hématite formée durant le stade d'apogée de diagenèse, et que l'aimantation C est portée par l'hématite formée subséquemment durant le stade d'altération hydrothermale à haute température. Les aimantations D incohérentes résultent de la dégradation de l'hématite en goethite, créée par la pénétration des eaux météoriques de basse température le long des zones de failles qui ont été continuellement réactivées depuis le Précambrien tardif. Les valeurs de $\delta^{18}O$ des minéraux argileux et de l'hématite contemporaine qui porte l'aimantation B et C indiquent que ces minéraux ont cristallisé à partir d'un fluide dont les températures étaient comprises entre 150 et 200°C, et avec les valeurs de $\delta^{18}O$ près de 1,0‰. Les fluides qui ont produit l'hématite précoce portant l'aimantation A sont relativement appauvris en ^{18}O, leurs valeurs approximatives sont de 0,8‰, et leur température était légèrement inférieure, soit 120–160°C. L'entremêlement des aimantations A, B et C indique que la précipitation de l'hématite provenait soit d'un fluide commun et avec reprécipitation par un fluide subséquent, ou de l'écoulement d'un fluide subissant des variations locales de perméabilité. Évidemment, cet écoulement du fluide était épidosique, à l'échelle du bassin, et il a été maintenu durant un laps de temps de l'ordre de 10^8 années. Il est suggéré que la stratigraphie des grès contrôlaient la migration latérale des fluides à la grandeur du bassin, et que les failles facilitaient l'écoulement des fluides d'une formation à l'autre.

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Introduction

Description of the fluid regimes that produce diagenetic mineral phases in sedimentary basins requires estimation of the temperature, age, and origin of the fluids. Because iron oxides are common products of diagenesis, and are capable of becoming magnetized by chemical remanent magnetization (CRM) when deposited at temperatures below their Curie temperatures, paleomagnetism can provide critical information about the age of fluids. For example, if several distinct diagenetically produced magnetizations were observed in a basin, they would signify that several fluid events had occurred. Furthermore, if these magnetizations can be compared with those in rocks whose ages are known, estimates can be made of the ages of diagenesis and of the fluid events associated with diagenesis. The objective of this work has been to make paleomagnetic age estimates of diagenetic hematite in the Athabasca Basin and then to compare them with radiometric ages obtained from coexisting illite and uraninite whose temperature and fluid-derived history can be documented from isotopic studies.

In the Athabasca Basin, fluid-inclusion microthermometry, isotope geochemistry, and studies of petrographic relationships among coexisting mineral phases have indicated that the sediments have been affected by several discrete fluid events that started in the mid-Proterozoic and continued to Recent times (Pagel *et al.* 1980; Wallis *et al.* 1983; Wilson and Kyser 1987; Kotzer and Kyser 1990*a*, 1990*b*). The earlier events involved high-grade diagenesis (prograde diagenesis; Dunoyer de Segonzac 1970) by isotopically and chemically evolved sedimentary brines at burial temperatures of approximately 200°C. This was followed by later, low-grade diagenesis (retrograde diagenesis; Dunoyer de Segonzac 1970), involving low-temperature meteoric waters at temperatures of <100°C (Kotzer and Kyser 1990*a*). Isotope studies also indicate that during both earlier and later diagenesis, temporally and isotopically distinct fluids occurred on a basin-wide scale (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*, 1990*b*). In the Athabasca Basin, only the prograde diagenetic events produced interpretable magnetizations. Magnetizations produced during the later retrograde diagenesis are very weak and incoherent in direction.

The origin and initial composition of the fluids that were responsible for prograde diagenesis in the Athabasca Basin have been isotopically and chemically modified by interactions at elevated temperatures with the sediments in the basin. In this paper, saline sedimentary brines are referred to as basin fluids. They were responsible for the deposition of many of the diagenetic minerals. These fluids may have been ancient meteoric water deposited contemporaneously with the sediments and modified chemically and isotopically via water–rock interactions during burial. It is also possible that the basin fluids migrated from adjacent orogenic belts as gravity-driven fluids flowing from higher elevations (McCabe *et al.* 1984; Oliver 1986; Miller and Kent 1988; McCabe and Elmore 1989). A second fluid, referred to as a basement fluid, which is isotopically distinct from the basin fluid, is thought to have issued upwards near and across the unconformity between the basement rocks and the Athabasca sediments along fault zones at various times in the Proterozoic and mixed with basin fluids to form uranium deposits (Wilson and Kyser 1987; Kotzer and Kyser 1991). These fault zones in the Athabasca Basin are rooted in the metasedimentary rocks of the basement. The basement fluid was responsible for Mg-chlorite formation in the altered metasedimentary basement rocks near fault zones (Wilson and Kyser 1987). The basement fluid may have originated as a byproduct of metamorphism generated by dewatering of the metasedimentary rocks during consolidation of the basement crust following closure of the Trans-Hudson Orogen. More likely, the basement fluids originated as sea water or ancient meteoric waters trapped in the basement rocks during uplift before the basin formed and later modified by subsequent water – basement rock interactions over a long period of time (Kyser and Kerrich 1990). Mechanisms for the generation of these types of fluids are considered to be analagous to those thought to occur in crystalline rocks of the Precambrian Shield (Frape and Fritz 1987).

Diagenetic hematite formed throughout the basin has a varying paragenesis as a result of reactions between oxidizing basin fluids and detrital Fe–Ti oxides and silicates. In some instances hematite is intimately associated with uranium ore formation; in others it is associated with quartz overgrowths formed during the earliest phases of diagenesis; and in yet others it occurs with euhedral quartz, illite, and uranium formed at peak burial temperatures. Hematite also occurs sporadically with goethite, infilling very late fractures. Hence, it is likely to have formed at several different times over a long interval.

Paleomagnetism has been used as a means of estimating the age of authigenic magnetic minerals (e.g., Elmore *et al.* 1985). Secondary authigenic hematite in red beds carries a CRM which, under favorable circumstances, can be separated from the original depositional remanence (DRM) and assigned ages (Roy and Park 1974; Roy and Robertson 1978; McCabe and Van der Voo 1983; Irving and Strong 1984; Bethke and Marshak 1990; Witte and Kent 1990). Secondary authigenic magnetite in carbonates may carry a CRM appropriate to the time of fluid migration and diagenesis (McCabe *et al.* 1984; Lu *et al.* 1990). Secondary hematite or magnetite in igneous rocks can carry a CRM dateable by these means (Irving and Strong 1985; Globerman and Irving 1988). Recently, paleomagnetic methods have been employed to estimate the age of fluids that produced Mississippi Valley-type ore deposits (Pan *et al.* 1990).

Previous paleomagnetic studies of hematitic sedimentary rocks in drill core from the Athabasca Basin yielded stable remanences (Fahrig *et al.* 1978; Ramaekers 1978). Interpretation was hampered by the core samples not being azimuth oriented and by the lack of knowledge at the time of fluid events within the basin. These data are discussed later.

In this study, oriented samples were collected from places where the time–temperature–fluid histories had been previously documented from isotopic, chemical, and petrographic studies. The magnetizations were first measured, and then the oxygen isotopic compositions of the hematite and coeval clay minerals in the same samples were determined and used to characterize the temperature, isotopic composition, and origin of the fluids from which the hematite was deposited. In this way, relationships between observed magnetizations and isotopically characterized fluid events could be established.

Geology, alteration, and age relations

The Athabasca Basin, located in the Churchill Province of northern Saskatchewan (Fig. 1), is the youngest of a series of intracratonic basins that formed following the Hudsonian orogeny (Ramaekers 1981). The fill consists of a thick sequence of mature quartz sandstones, marine shales, and shelf car-

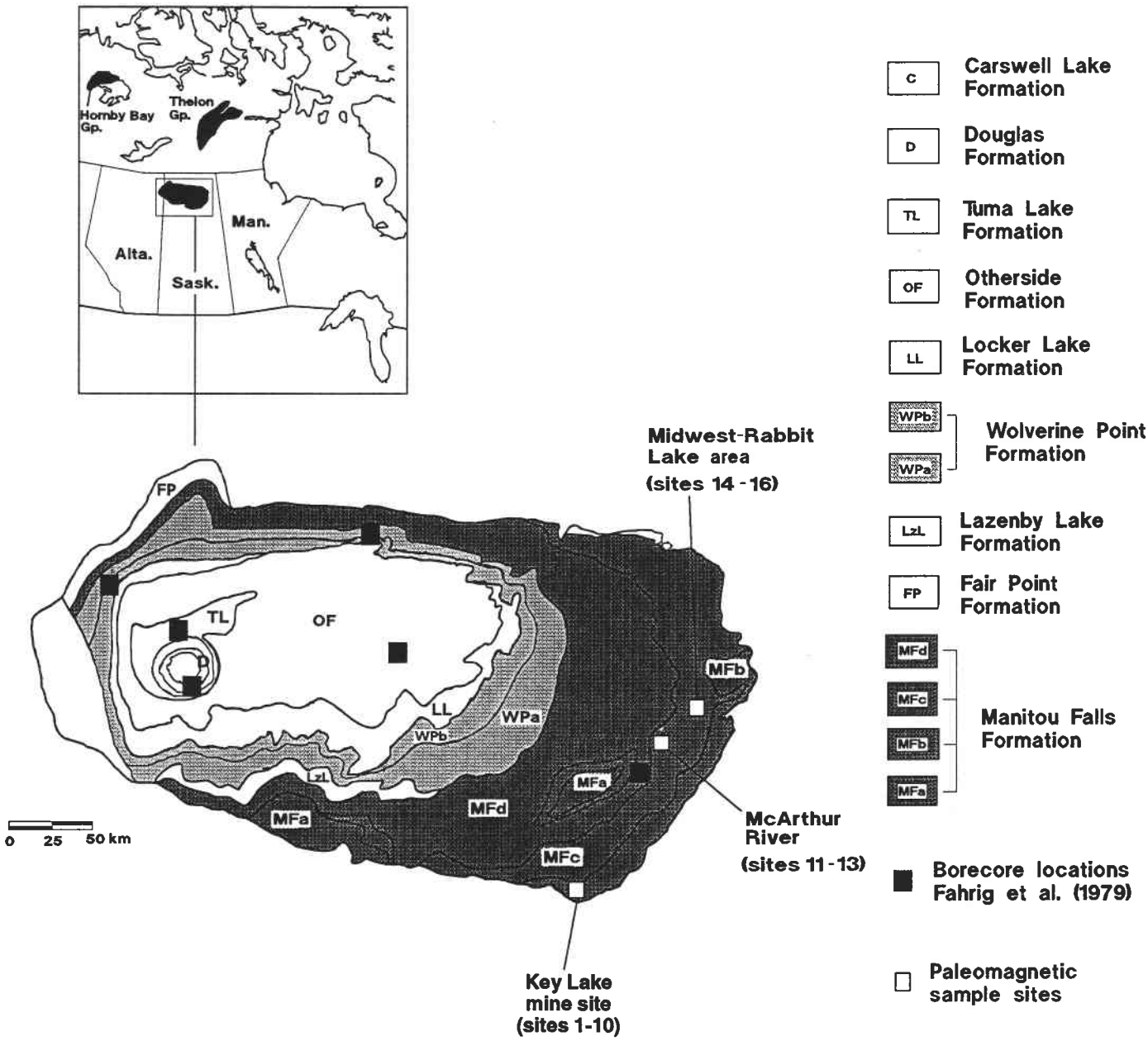


FIG. 1. Geologic map of Proterozoic Athabasca Basin showing locations of paleomagnetic samples collected from outcrop (this study) and drill core (Fahrig et al. 1978). Also shown is generalized stratigraphy of Athabasca sediments (from Ramaekers 1981).

bonates. They unconformably overlie Archean and Early Proterozoic gneisses and metasedimentary rocks (Ramaekers 1979). At the unconformity, there is a well-developed hematitic paleoregolith several metres thick (Tremblay 1982).

The time of original deposition is uncertain. From radiometric studies of diagenetic minerals, both in the sedimentary rocks and in the underlying metasediments, a minimum age of between approximately 1500 and 1720 Ma has been estimated (Fahrig *et al.* 1978; Ramaekers 1981; Armstrong and Ramaekers 1985; Cummings *et al.* 1987; Kotzer and Kyser 1990*a*; Machado 1990).

During the mid-Proterozoic, high-grade U–Ni–Co–As–Cu–Zn deposits formed in fault zones that intersect the basal unconformity (Hoeve and Quirt 1984). Oxidizing, high-temperature (200°C), saline (30–34 wt.% NaCl) basin fluids leached metals from the sediments, mixed with reducing basement fluids, and deposited the metals in the fault zones (Hoeve and Sibbald 1978; Wilson and Kyser 1987; Kotzer and Kyser 1990*a*). Mixing of these two isotopically and chemically distinct fluids is recorded by the occurrence of quartz, illite, kaolinite, tourmaline, chlorite, and hematite, which have formed alteration haloes peripheral to the uranium deposits. Outside these alteration haloes, the sandstones contain an assemblage of illite, kaolinite, and hematite. Petrographic relations and isotopic and chemical compositions of various mineral phases have been used (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*) to define the paragenesis of diagenetic minerals in the basin (Fig. 2).

Uplift began in the late Precambrian and has continued intermittently to Recent times (Hoeve and Quirt 1984). This resulted in reactivation of faults and erosion of the basin fill. It also resulted in the incursion of mid- to high-latitude, low-

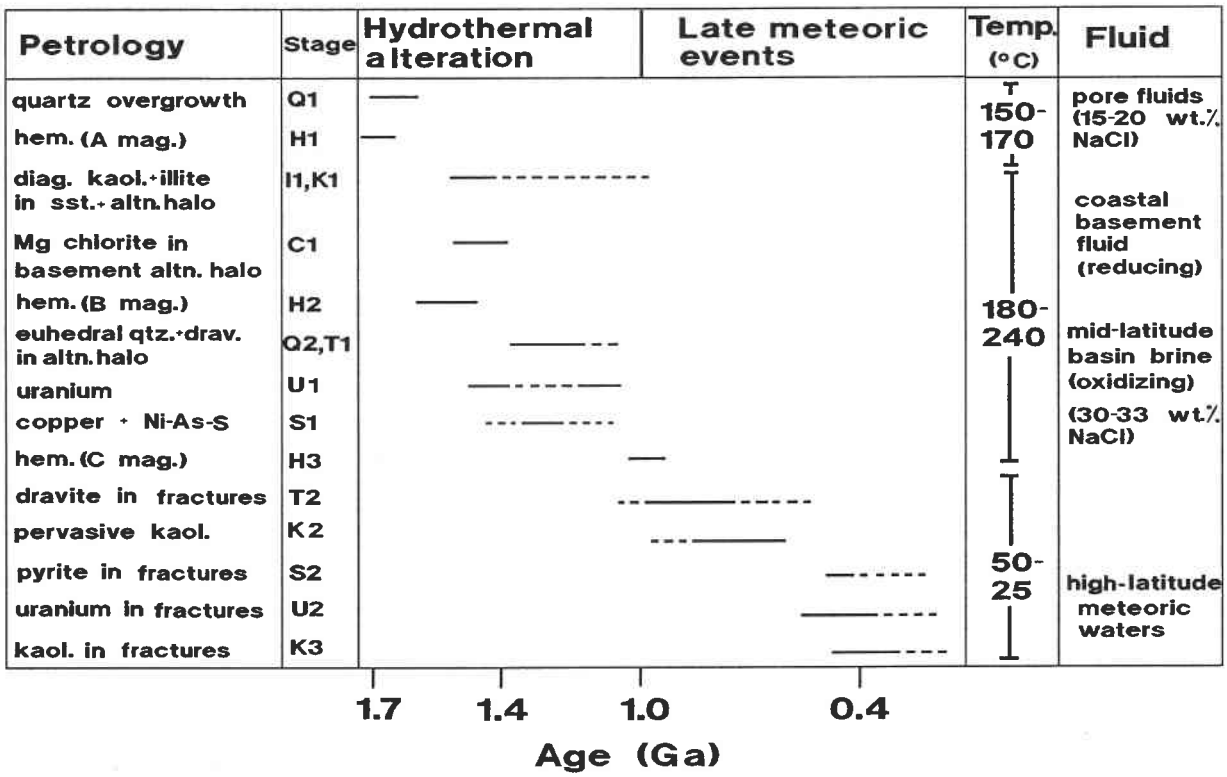


FIG. 2. Proposed fluid − mineral age paragenesis for various minerals in the Athabasca Basin developed from petrographic, fluid-inclusion, and stable and radiogenic isotope systematics. Paragenesis of hematite has been established using field and petrographic relationships and paleomagnetism (modified from Kotzer and Kyser 1990*a*). altn., alteration; drav., dravite; hem., hematite; kaol., kaolinite; sst., sandstone; qtz., quartz.

temperature meteoric waters along faults and the formation of kaolinite and sporadic goethite together with altered illite, chlorite, and uranium minerals (Wilson and Kyser 1987; Kotzer and Kyser 1991).

The Athabasca Basin has had a complex fluid history as reflected by the wide age ranges for the various diagenetic minerals. U−Pb ages of uraninite from high-grade uranium mineralization range widely from 1521 ± 8 to 903 ± 40 Ma (Hoehndorf *et al.* 1985; Ruhrmann and Von Pechmann 1989; Carl *et al.* 1988; Cummings and Krstic 1991). $\delta^{18}O$ values as low as −27‰ and reset U/Pb ages as young as 200 Ma in the uraninites (Hoehndorf *et al.* 1985; Kyser *et al.* 1990) reflect alteration of uranium ores by late meteoric waters. High-temperature, diagenetic illites throughout most of the basin, which have $\delta^{18}O$ and δD values and crystallinity indices indicating formation from high-temperature basin fluids, yield an Rb−Sr isochron age of 1477 ± 57 Ma, although a few illites in sandstones near Rabbit Lake have model Rb−Sr ages near 900 Ma (Kotzer and Kyser 1990*b*). K−Ar ages of illites with similar paragenesis range from 1500 to 400 Ma, the younger ages being attributed to resetting of the K−Ar systematics by late-stage, meteoric waters (Wilson *et al.* 1987; Kotzer and Kyser 1990*b*, 1991). Evidently the history of fluid flow in the Athabasca Basin has been long and protracted and includes several high-temperature (200°C) events that occurred during the Proterozoic and one or more low-temperature (<100°C) Phanerozoic events.

Sampling and experimental procedures

Samples were collected from sites where previous isotopic and geochemical work revealed evidence of high-temperature fluid flow and resultant diagenesis (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*). In addition, some samples were obtained from sites at which there was evidence of the effects of both high-temperature, diagenetic basin fluids and later low-temperature meteoric water.

All samples were from the Manitou Falls Formation. They were collected from the open-pit mine at Key Lake (sites 1−10 in Fig. 1 and Table 1) and from exposed outcrops at McArthur River (sites 11−13) and Midwest Lake − Rabbit Lake (sites 14−16). Samples from Key Lake were all within 40 m of the unconformity. At Midwest Lake − Rabbit Lake and McArthur River they were much higher in the stratigraphic sequence, approximately 100 and 400 m above the unconformity, respectively. All samples were oriented using both a magnetic and sun compass, and at each site five to nine drill cores or a few hand specimens were taken. Hand samples were cored in the laboratory, and two cylindrical specimens (2.5 cm diameter and 2.2 cm length) were cut from each core.

The natural remanent magnetization (NRM) of the cores was measured using an automated Schoenstedt SSM1 fluxgate spinner magnetometer. The smallest magnetization that can be routinely measured by such instruments is approximately 0.5×10^{-3} A/m. Data from the analyses were treated using Fisher statistics (Fisher 1953), and results were plotted on stereograms and orthogonal vector diagrams (Zijderveld 1967). Pilot specimens (usually a pair of specimens cut from the same core) were subjected to alternating-field (AF) and thermal demagnetization. AF's as high as 180 mT and temperatures up to 750°C were used.

Interstitial clay minerals and hematite were separated from oriented drill cores that had yielded measureable NRM's. The

TABLE 1. Description of collection sites (see Fig. 1)

Site	Description
1	MFa; Deihlmann Pit – Key Lake; altered (kaolinitic) sandstone, 20 m from Key Lake fault
2*	MFa; Deihlmann Pit – Key Lake; purple, hematitic bands in altered sandstone, 100 m from Key Lake fault
3	MFa; Gaertner Pit – Key Lake; highly bleached sandstone near axis of Key Lake fault
4	MFa; Gaertner Pit – Key Lake; silicified, altered sandstone, very pale red staining
5	MFa; Gaertner Pit – Key Lake; similar to site 4 (2 m above).
6*	MFa; Gaertner Pit – Key Lake; reddish hematitic sandstone, ~20 m from Key Lake fault
7	MFa; Gaertner Pit – Key Lake; bleached sandstone, some thin hematite layers (2 m above site 6)
8	MFa; Gaertner Pit – Key Lake; very white, altered sandstone
9*	Wollaston Group metasediments; Gaertner Pit – Key Lake; hematitic, relatively unaltered paleoregolith (5 m below unconformity)
10	Wollaston Group metasediments; Gaertner Pit – Key Lake; altered chloritic paleoregolith below uranium mineralization, 3 m below unconformity
11	MFc; McArthur River area; silicified kaolinitized sandstone
12	MFb–MFc; McArthur River (Whitford Lake area); silicified, hematitic, highly brecciated sandstone
13	MFb; McArthur River (Whitford Lake area); silicified, brecciated, hematitic sandstone
14*	MFb; Midwest Lake Road; sandstone with 5–10 cm thick, purple hematite bands showing some later bleaching effects
15*	MFb; Midwest Lake Road; sandstone with purple hematite bands crosscut by reddish hematite (liesegang rings)
16*	MFb; Midwest Lake Road; sandstone with grey-purple hematite and very little bleaching

NOTES: MFa, MFb, and MFc, Manitou Falls Formation (A, B, and C groups). Bedding attitude for site 3 is 025°/6° SE; all other sites are flat lying.
*Site yielding measureable and coherent NRM's.

<2 μm size fraction of the clay minerals used for isotopic analysis was obtained by crushing and centrifuging. Modal mineralogies of the clay separates were determined by semi-quantitative X-ray diffraction (XRD) analysis. Pure hematite was obtained from drill cores using a combination of crushing, centrifuging, magnetic separation, and mild acid treatments (dilute HF) to remove intergrown clay minerals. Purity of the hematite separates was checked by optical examination and XRD.

Oxygen isotopic compositions of the clay minerals and the hematite were determined using the BrF_5 technique of Clayton and Mayeda (1963). The hydrogen-isotope compositions of clay minerals were determined using the technique of Bigeleisen *et al.* (1952) as modified by Kyser and O'Neil (1984). All stable isotope measurements were made using a MAT 251 mass spectrometer and are reported using the δ notation in units of per mil (‰) relative to the standard standard mean ocean water (V-SMOW). Replicate $\delta^{18}O$ analyses of the clay minerals and hematite are reproducible to ± 0.2‰ and δD values to ± 3.0‰. Using the above procedures, the $\delta^{18}O$ value of National Bureau of Standards (NBS) 28 quartz standard is 9.6‰, and the δD value of NBS 30 biotite standard is -65‰.

The oxygen-isotope fractionation factors used in this paper are those suggested by Eslinger and Savin (1973) for illite–water and by Land and Dutton (1978) for kaolinite–water. The hematite–water fractionations are assumed to be similar to the magnetite–water fractionations generated by Becker and Clayton (1976).

Occurrence of hematite

Hematite occurs throughout the basin, most commonly in the basal part of the Manitou Falls Formation (Fig. 1). It imparts a dark purple to reddish color to the sandstones. In the Key Lake area, fine-grained hematite and specular hematite occur intergrown with early quartz overgrowths, as interstitial minerals in dark purple sandstones, and in reddish illite-rich alteration haloes surrounding uranium ore (Fig. 2). In the Midwest Lake – Rabbit Lake area, specular hematite occurs within the dark purple sandstones and in crosscutting reddish liesegang rings.

Scanning electron microscope (SEM) studies reveal that hematite, with negligible amounts of magnetite, is the main iron oxide mineral in all samples of sandstone and paleoregolith. Hematitic sandstones and paleoregolith from the Key Lake area (sites 2, 6, 9) contain hematite of varying size and paragenesis. The dark purple, banded sandstones located about 100 m from the main fault zone at Key Lake (site 2) contain specular hematite and rutile which has replaced detrital ilmenite intergrown with interstitial clay minerals (Fig. 3*a*). Occasionally, dark colored, framboidal hematite is replaced by laths of highly reflective, specular hematite (Fig. 3*b*), indicating that there was more than one stage of hematite formation. Samples of the reddish, hematitic sandstones located about 20 m from the main fault zone at Key Lake (site 6) contain dominantly interstitial illite and highly reflective, specular hematite and rutile (Fig. 3*c*). Electron microprobe analysis of hematite in these samples indicates as much as 10% Ti. Samples of sandstones collected from outcrops in the Midwest Lake – Rabbit Lake area (sites 14–16) were taken from stratigraphically controlled hematite horizons and from leisegang rings. Polished thin sections display minor amounts of specular hematite–rutile intergrown with illite and kaolinite in highly silicified sandstones (Fig. 3*d*).

Paleomagnetic results

Four magnetic types were recognized. A has southeast directions with positive inclinations, B has north-northeast directions with negative inclinations, and C has steep, positive inclinations (Fig. 4). D has widely scattered directions and very low intensities.

Six sites from Key Lake and the Midwest Lake – Rabbit Lake area contained NRM's with intensities between 9 and 0.5×10^{-3} A/m. AF demagnetization of pilot specimens generally removed very little of the NRM, other than a small, recently acquired viscous remanent magnetization (VRM) having low coercivities. AF demagnetization was incapable of resolving magnetic components. Thermal demagnetization of

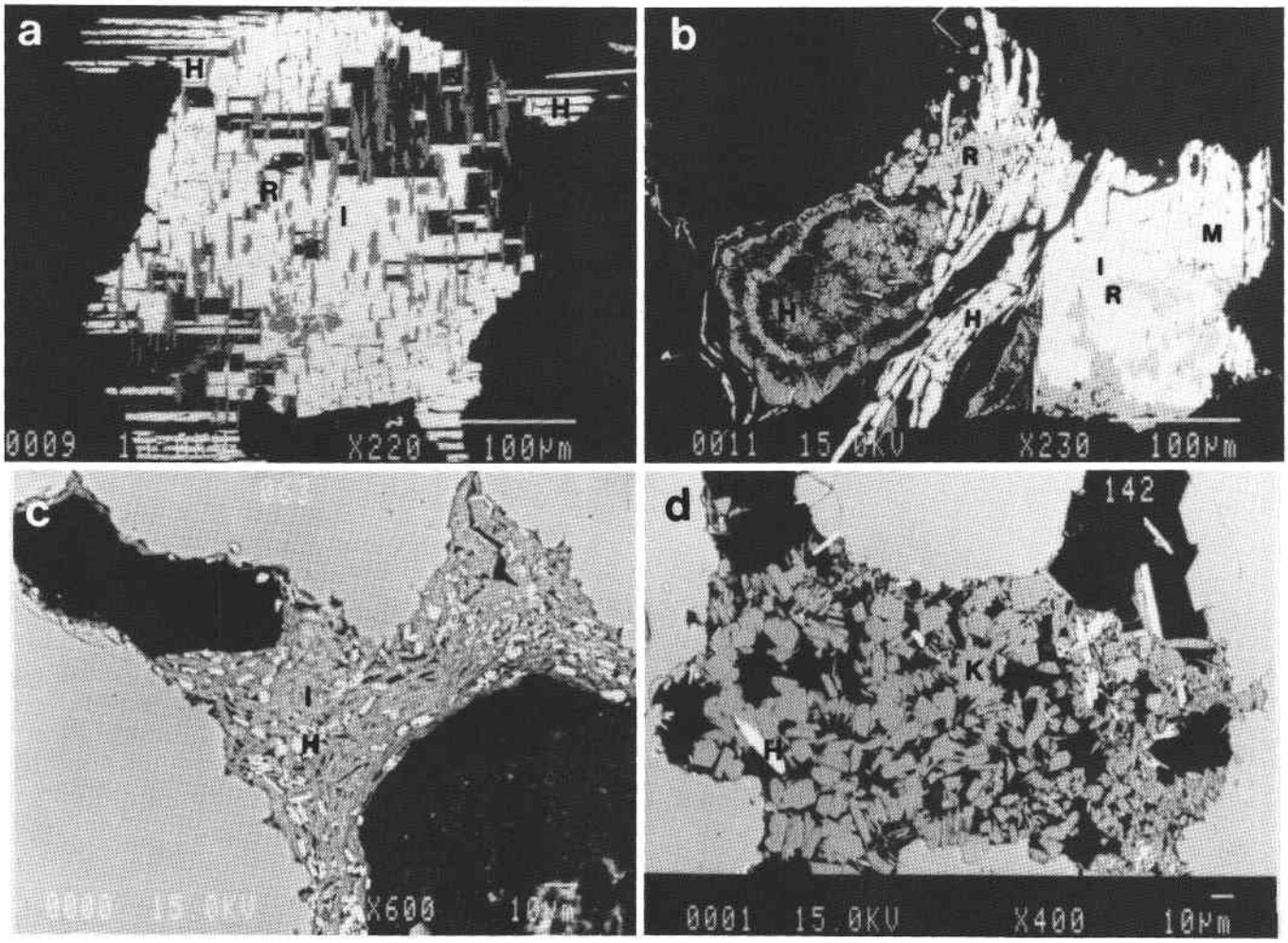


FIG. 3. Photomicrographs of interstitial Fe–Ti oxides and clay minerals from sample sites examined in this study, illustrating the deposition of iron oxide and clay minerals from high-temperature diagenetic fluids as well as complex intergrowths of hematite and Fe–Ti oxide minerals. (*a*) Ilmenite (I) replaced by hematite (H) and exsolved rutile (R) during early diagenesis of heavy mineral bands in sandstones at site 2. Dominant A and subordinate C magnetizations are present. (*b*) Complex intergrowth of hematite (H), rutile (R), magnetite (M), and ilmenite (I) in dark purple hematite bands at site 2. Presence of multiple stages of hematite (dark, framboidal and light, lathlike), indicating that more than one diagenetic fluid event has occurred resulting in formation of hematite carrying either A or C magnetization. (*c*) Interstitial illite (I) and hematite (H) carrying B magnetization in illite alteration haloes near uranium mineralization at Key Lake (site 6). (*d*) Interstitial hematite (H) carrying C magnetization intergrown with kaolinite (K) in dark purple heavy mineral bands at the Midwest Lake – Rabbit Lake outcrops (site 14).

pilot specimens (100°C increments to 400°C and 50°C increments from 400 to 700°C) generally yielded a remanent magnetism that was consistent in direction and exhibited a linear decay to the origin on orthogonal plots (Fig. 5). Only single magnetizations are present. The intensities of magnetization dropped rapidly above temperatures of 600°C and especially between 650 and 700°C. This combination of high unblocking temperatures and high coercivities indicates that the NRM resides predominantly in hematite (Neel temperature 675°C). On the basis of these pilot results, all remaining samples were demagnetized using a minimum of five steps between 400 and 700°C. During demagnetization, the samples yielded consistent directions of magnetization (sites 2, 6, 9, 14, 15, and 16 in Table 2). All samples with A, B, and C directions exhibited these characteristics.

Samples altered by late meteoric water at Key Lake and McArthur River (sites 1, 3, 4, 5, 7, 8, 10, 11, 12, and 13) exhibited different magnetic behavior. These specimens showed faint traces of hematite and fracture-filling goethite and contained a significant component of Mg-chlorite or very fine-grained (and probably paramagnetic) hematite. NRM intensities were less than 1×10^{-3} A/m and during demagnetization fell below the limit of detection. The directions are scattered, and demagnetization does not resolve any components. They are referred to as D magnetizations. They provide no record of the paleofield. Their incoherent nature and the presence of goethite indicates that the late-stage alteration occurred at temperatures less than 80°C because above this temperature goethite transforms to hematite and acquires a remanence (Hodych *et al.* 1984). The low NRM intensities of the D magnetization may make the samples more amenable to further magnetic studies using a cryogenic magnetometer, which has detection limits approximately two to three orders of magnitude lower than the magnetometer used in this study.

Key Lake mine site

A, B and C magnetizations are present (Fig. 4*a*). Specimens with C magnetizations are relatively few. Sandstones with

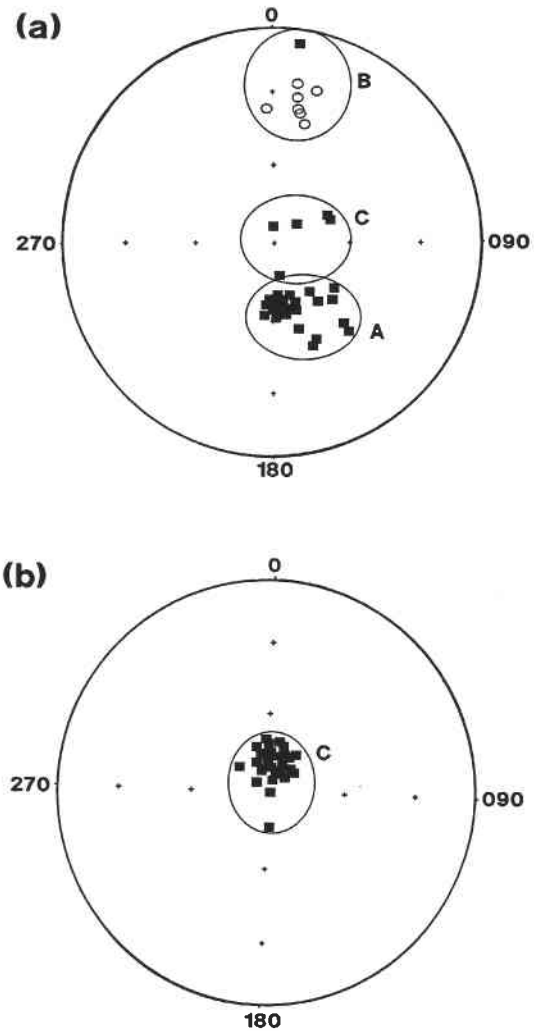


FIG. 4. Initial NRM directions. (*a*) Samples from Key Lake mine site (sites 2, 6, 9). (*b*) Natural outcrop samples from Midwest Lake – Rabbit Lake area (sites 14, 15, 16). ■ and ○, positive and negative inclinations signifying normal and reversed polarity, respectively.

dark-purple, hematite horizons within 100 m of the fault zones (site 2) have A magnetization (Table 2; Fig. 5*a*). C magnetizations occurred in three samples from site 2. The hematitic paleoregolith beneath the Athabasca sandstones (site 9) has a dominantly A magnetization (Table 2; Fig. 5*b*) and C magnetization occurs in two samples. Data from sites 2 and 9 have been combined to give an overall mean for the A magnetization (Fig. 6*a*; Table 2). When calculating this and other mean values of Table 2, individual specimens have been lumped together and given unit weight. This is justifiable on the grounds that the CRM process has, in all probability, integrated the field over hundreds to thousands of years, intervals that differ from specimen to specimen at the same collecting site. As the formation of diagenetic hematite occurs over a similar time interval, secular variations in the magnetic field are likely to be averaged among the samples.

Hematitic sandstone located within 20 m of the main fault zone and containing high-grade uranium mineralization (site 6) carries B magnetizations that have negative inclinations (Table 2; Figs. 5*c*, 6*b*). One sample from site 6 contains two-component magnetization (Fig. 5*d*). During demagnetization, a northerly directed magnetization, with shallow, normal inclination (declination $D = 13°$, inclination $I = 4°$; Fig. 5*d*) is removed between 100 and 450°C. Further demagnetization to temperatures up to 700°C yielded a northerly directed magnetization, with a shallow, reversed inclination ($D = 14°$, $I = -3°$; Fig. 5*d*). Such behavior occurred in only one sample, and its significance is unclear.

Midwest Lake – Rabbit Lake area

All sites are located within approximately 300 m of each other (14, 15, 16) and yield C magnetizations (Fig. 4*b*). The directions are not far from the present Earth's field (PEF). However, above temperatures of 100°C, there was no significant change in the direction and little change in intensity of the magnetization up to 650°C, and then linear decay to the origin on orthogonal plots (Fig. 5*e*). Evidently the magnetization is not a recent VRM acquired in the PEF. The C magnetizations from here and from Key Lake have been combined to yield an overall mean direction (Table 2; Fig. 6*c*).

Thermal demagnetization curves

Representative thermal decay curves are shown in Fig. 7. In some samples, there is a slight change in the magnetic intensity between temperatures of 100 and 500°C, probably due to removal of very small, recently acquired VRM's. There is a marked decrease of 10–25% in the magnetic intensity between 500 and 600°C. SEM and electron microprobe analyses reveal the presence of titaniferous hematite that was formed by the breakdown of detrital ilmenite during diagenesis. Titanium depresses the Neel temperature of hematite and could be reponsible for the decrease in intensity between 500 and 600°C. Remnants of magnetite that had been incompletely altered to hematite could also be responsible; however, only negligible amounts of magnetite have been observed. The most dramatic decrease, however, is between 670 and 700°C as the magnetization carried by essentially pure hematite is removed. Both the lower and higher unblocking temperature magnetizations have the same direction of magnetization. Hence, the mineral phases carrying the two magnetizations probably were formed during the same diagenetic event.

Isotopic composition of hematite, coexisting clay minerals, and formation fluids

Stable-isotope (O, H) and mineralogic studies of clay minerals in the sandstones indicate that most have formed from a reducing-basement fluid or an oxidizing-basin fluid having δD values of approximately −20‰ and −50‰, respectively (Wilson and Kyser 1987). From stable-isotope (O, H) analyses of fluids in inclusions in euhedral quartz, and of coexisting clay minerals and quartz, the diagenetic basin fluids have calculated $\delta^{18}O$ and measured δD values of 3 ± 3‰ and −50 to −60‰, respectively (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*). Oxygen and hydrogen isotopic analyses of clay minerals produced by late retrograde alteration indicate that these fluids had $\delta^{18}O$ and δD values of approximately −15 to −20‰ and −120 to −130‰, respectively, values similar to modern-day meteoric waters in the Athabasca Basin.

Detailed paragenetic and stable isotopic studies of the diagenetic minerals indicate that (*i*) the basin fluids responsible for the diagenetic illite, kaolinite, and quartz which are coeval with hematite carrying B and C magnetizations had temperatures of 180–220°C and $\delta^{18}O$ values of 3 ± 3‰;

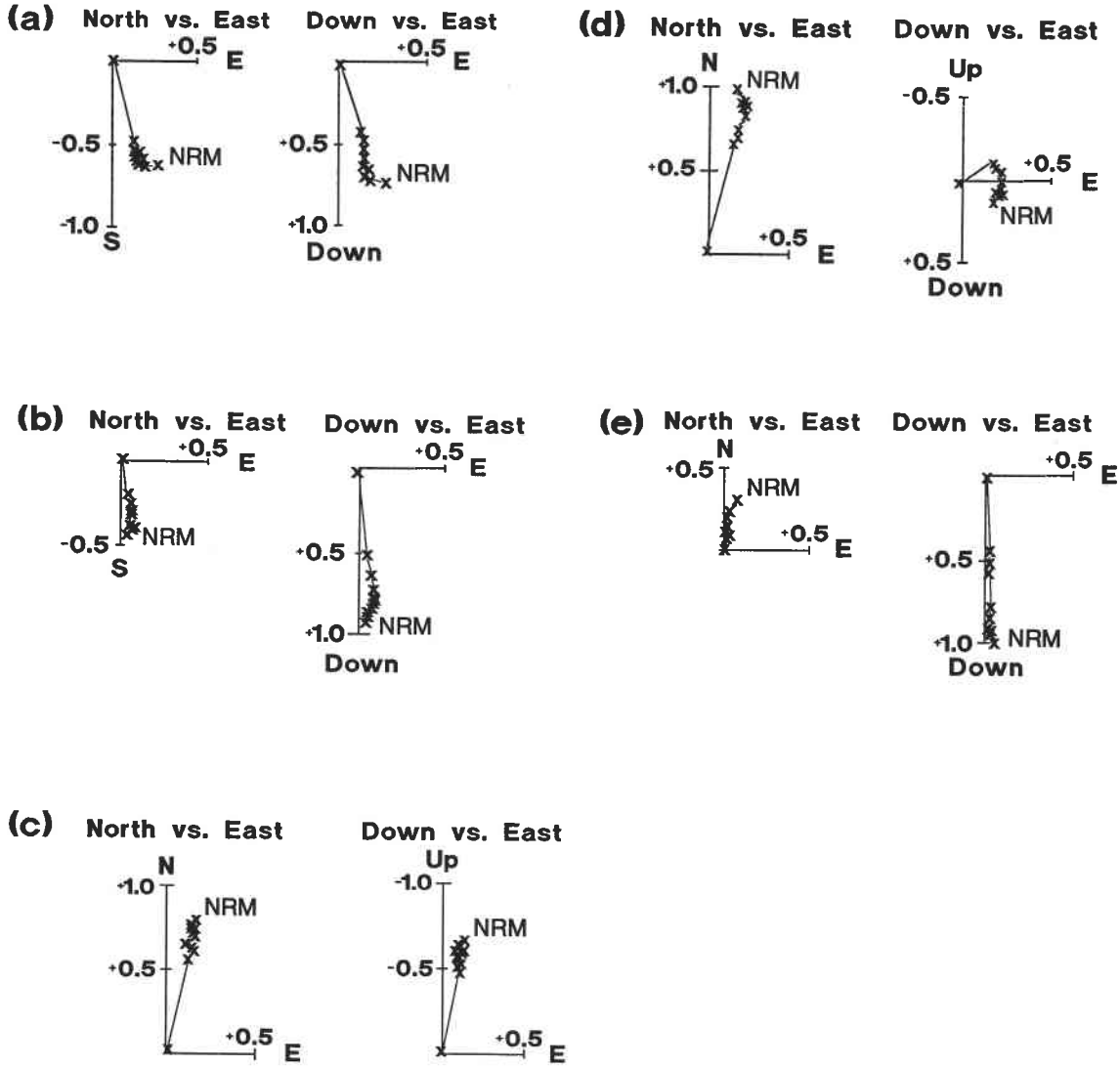


FIG. 5. Orthogonal diagrams of specimens from Key Lake mine site and Midwest Lake – Rabbit Lake sites, illustrating typical behavior during thermal demagnetization at temperatures up to 700°C. NRM, natural remanent magnetization. (*a*) Dark purple, banded sandstones with specular hematite distal to fault zone at Key Lake (site 2). (*b*) Hematitic paleoregolith approximately 5–10 m below unconformity at Key Lake (site 9). (*c*) Red sandstones with specular hematite peripheral to fault zones hosting uranium deposits at Key Lake (single component, site 6). (*d*) Red sandstones with specular hematite peripheral to fault zones hosting uranium at Key Lake (two component, site 6). (*e*) Specular hematite in dark purple, banded sandstones and liesegang rings from Midwest Lake – Rabbit Lake area (site 14).

and (*ii*) the fluids that deposited the early quartz overgrowths intergrown with hematite carrying A magnetizations had lower temperatures of 120–150°C and lower $\delta^{18}O$ values near 0‰ (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*, 1990*b*). Comparable isotopic studies on the hematite responsible for the magnetizations observed are now discussed.

Examination of polished thin sections indicates that the hematite carrying the measureable magnetization was coeval with clay minerals (Figs. 3*c*, 3*d*). The $\delta^{18}O$ values of the hematite responsible for the A, B and C magnetizations range from −6.4 to −7.4‰ (Table 3). Illite–magnetite and kaolinite–magnetite geothermometry, using the $\delta^{18}O$ values for hematite and coexisting kaolinite and illite in samples with B and C magnetizations, indicate that they formed from fluids with temperatures of 140–180°C and $\delta^{18}O$ values of 0.1–1.2‰ (Table 3; Figs. 8*a*, 8*b*). Quartz-magnetite geothermometry, using the $\delta^{18}O$ values of hematite and coeval quartz overgrowth obtained from samples with A magnetization, indicates that they have formed from a fluid having temperatures of approximately 150–160°C and $\delta^{18}O$ values of approximately 0.7–0.9‰ (Table 3; Figs. 8*a*, 8*b*). The temperatures and $\delta^{18}O$ values of the fluids from which the generations of hematite formed are comparable to those obtained previously from clay minerals and silicates (Fig. 8*b*), and demonstrate that the paleomagnetic information from the hematite can be directly linked to the isotopic observations and to genetic inferences from them.

Ages of magnetization

In principle, the ages of Athabascan magnetizations can be established by matching them with magnetizations (or corresponding paleopoles) observed in well-dated Proterozoic rocks. Such age estimates should then agree with the radiomet-

TABLE 2. Summary of paleomagnetic data from Athabasca Basin

Site no.	N	n	D (°)	I (°)	Type	α_{95} (°)	k	Demagnetization range represented
				KEY LAKE MINE SITE				
2	5	9	159	53	A	8	40	450–650°C, 100 mT
2	1	1	338	80	C			450–650°C, —
2	1	1	066	69	C			450–650°C, —
6	3	6	010	−38	B	6	118	450–650°C, 100 mT
9	6	12	163	68	A	2	334	450–650°C, 100 mT
9	1	2	065	69	C			450–650°C, —
				MIDWEST LAKE–RABBIT LAKE AREA				
14	5	10	005	79	C	2	467	450–650°C, 100 mT
15	6	8	031	82	C	6	90	450–650°C, 100 mT
16	5	9	006	76	C	2	872	450–650°C, —

A magnetization
Mean: $D = 158°$, $I = 62°$, $\alpha_{95} = 5°$, $k = 48$, $n = 21$, $\lambda = 43°$
Pole: 12°N, 271°E, $d_p = 6°$, $d_m = 7°$
B magnetization
Mean: $D = 11°$, $I = -36°$, $\alpha_{95} = 7°$, $k = 76$, $n = 6$, $\lambda = 20°$
Pole: 12°S, 246°E, $d_p = 3°$, $d_m = 5°$
C magnetization
Mean: $D = 18°$, $I = 79°$, $\alpha_{95} = 3°$, $k = 95$, $n = 31$, $\lambda = 69°$
Pole: 77°N, 284°E, $d_p = 5°$, $d_m = 5°$

NOTES: N, number of cores; n, number of specimens measured; D and I, magnetic declination and inclination; Type, magnetization type; α_{95}, radius of the circle at the 95% confidence level; k, Fisher's precision; d_p and d_m, elliptical error in paleopole, radii of the ellipse of 95% confidence along and perpendicular to the direction of the magnetic meridian from the site to the pole, respectively; λ, paleolatitude.

ric ages determined on diagenetic minerals coeval with the magnetization-bearing hematite. Paleopoles presently available from the Laurentian Shield either differ in position from Athabascan paleopoles or, if they are close, are not well-dated. Age estimates made in this way can only fall within rather broad limits.

Relevant paleopoles are plotted in Fig. 9. Only those that have errors ($P = 0.05$) are considered. Paleopoles in the Pacific–American hemisphere, rather than their antipoles, are plotted and are assumed to be north poles following Irving and McGlynn (1976) and Hildebrand (1988). Magnetizations pointing towards (away from) these poles then have normal (reversed) polarity. On this basis, the A and C magnetizations are normal and B is reversed. All magnetization must postdate ages of approximately 1750–1790 Ma (Hohndorf *et al.* 1989; Annesley and Madore 1991) which define the time of waning stages of Hudsonian metamorphism in the basement rocks underlying the Athabasca sandstones.

A magnetization

The A magnetization is carried by hematite in the paleoregolith and by dark purple hematite in sandstones distal to fault zones at Key Lake. Petrographically, this hematite occurs with early quartz overgrowths (Fig. 2) and early oxidization of detrital heavy minerals formed during early basin diagenesis (Fig. 3*a*), and is therefore the oldest hematite in the basin. Apatite coeval with this earliest stage of diagenesis has yielded a U–Pb age of 1650–1700 Ma (Cummings *et al.* 1987).

The A paleopole falls in the midst of a broad east–west swath of paleopoles which stretches from the south of Mexico to the western Caribbean (Fig. 9). These paleopoles have often been loosely described as "post-Hudsonian." Like the A paleopole, all have normal polarity. Some are based on secondary post-folding magnetizations, such as those in sedimentary rocks from the Great Slave Supergroup (CO; Evans *et al.* 1980) and in the Eskimo volcanics of the Belcher Islands (EVO; Schmidt 1980). Others have been obtained from metamorphic rocks whose magnetizations have been acquired as a result of uplift and cooling following deformation, such as those in the metamorphic rocks of the Melville Peninsula and Daly Bay (MD; Park 1973) and the Itivdelq dykes and gneisses of Greenland (IT; Morgan 1976).

The age of post-Hudsonian paleopoles is poorly defined, but using as a guide Rb–Sr and K–Ar ages cited by authors for the metamorphic rocks that have yielded them, an Early Proterozoic age (1750–1600 Ma) is indicated. This is in good agreement with the U–Pb apatite age for the earliest stages of diagenesis in the Athabasca Basin cited above.

Post-Hudsonian paleopoles are dispersed mainly in an east–west sense. That they all have normal polarity indicates that the time span they record may be comparatively brief, shorter, that is, than the range quoted above. All have been calculated relative to present horizontal, so the effects of uncorrected tilts may be present. However, systematic tilts over so large a sampling area seem unlikely, and a more probable cause for their elongated distribution is relative rotations about vertical axes occurring after the main Hudsonian deformation. If this interpretation is correct, then it is notable that the four paleopoles from the Hudson Bay region (CS, EVO, MD, OI; Fig. 9) form the western part of the distribution, indicating a clockwise rotation of these localities relative to those from the central and western Laurentian Shield.

B magnetization

The B magnetization is found only in the reddish pervasive hematite and hematitic bands located close to the major fault

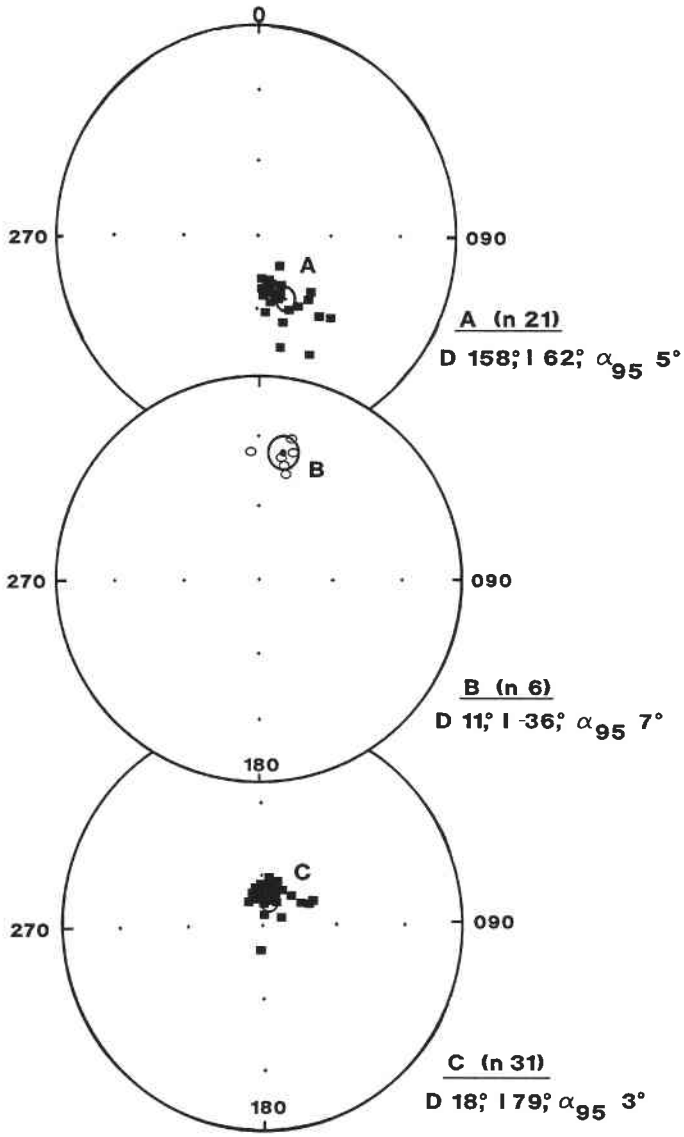


FIG. 6. Summary of directions in thermally demagnetized specimens from Key Lake and the Midwest Lake – Rabbit Lake area. (⊙), mean magnetic direction with error at 95% (α_{95}). Three separate, well-characterized magnetizations A, B, and C occur at Key Lake, whereas C magnetization predominates in the Midwest Lake – Rabbit Lake area. The three magnetizations can be correlated with the formation of three distinct phases of hematite. Symbols as in Fig. 4.

zones that host the uranium ore at Key Lake. An Rb–Sr age of 1477 ± 57 Ma (Kotzer and Kyser 1990*b*) has been obtained from illite intergrown with hematite carrying B magnetization (Fig. 3*c*).

The B magnetization is reversed. Its paleopole falls to the east of all Middle Proterozoic paleopoles from the Laurentian Shield (Fig. 9). Older Middle Proterozoic paleopoles (Fig. 9) have been derived from rocks with both normal and reversed polarity and span a time interval from about 1358 to 1480 Ma. Rock units represented are the Belt-Purcell Supergroup of the Rocky Mountains (BA, BM, BS), the Sibley Group of Ontario (SB), and anorthosite intrusions of Labrador (MA, ML, MI, MK, NC). Younger Middle Proterozoic and Upper Proterozoic paleopoles from Laurentia fall to the north and west

TABLE 3. Descriptions, mineralogical compositions determined from XRD, δD and $\delta^{18}O$ values of clay minerals and hematite, isotope equilibration temperatures, and $\delta^{18}O$ compositions of fluids from samples collected for paleomagnetic analyses

Site no.	Description and magnetic intensity	XRD results (%)	$\delta^{18}O$ (‰)		δD (‰)	Temp. (°C), Illite–, and kaolinite–hematite	$\delta^{18}O$ fluids (‰)
			Clay minerals	Hematite	Clay minerals		
2	Detrital ilmenite altered to hematite + rutile; two phases of hematite present; intergrown with illite; ~0.009 A/m	50 illite; 50 kaolinite	11	−6.4	−68	150–160	0.7–0.9
6	Altered quartz grains with illite + kaolinite; hematite + rutile in mixture of hematite–illite; ~0.05 A/m	>90 illite (trace kaolinite + dravite)	11.3	−7.4	−76	40–150	0.1
9	Altered biotite + quartz with hematite flakes and iron staining on biotite; ~0.009 A/m	illite–chlorite	10	—	−62	—	—
14	Quartz overgrowth with illite + kaolinite; very little pore space; ~0.07 A/m	30 illite; 70 kaolinite	11.7	—	−102	—	—
15	Quartz overgrowths with illite + hematite; ~0.07 A/m	10 illite; 90 kaolinite	—	—	—	—	—
16	Quartz overgrowths with illite + hematite; ~0.08 A/m	10 illite; 90 kaolinite	11.5	−6.9	−94	180–190	1.2

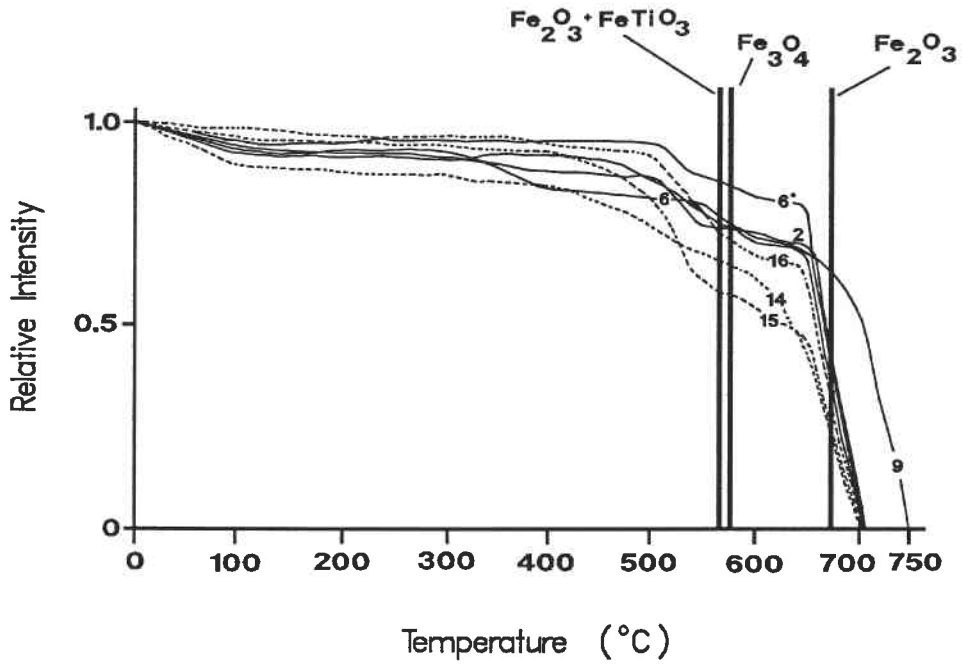


FIG. 7. Relative intensities of the magnetic components in the samples with temperature. Heavy vertical lines indicate Curie temperatures for Ti-rich hematite, magnetite, and hematite. Noticeable decrease in intensity between temperatures of 500 and 600°C with total loss of intensity between 650 and 700°C indicates that a small component of magnetite or Ti-hematite and a major component of hematite exist in all samples. Slight increases in intensity between temperatures of 100 and 500°C for some samples indicates removal of a recent viscous remanence or a detrital component. Numbers refer to the sample sites. *, two-component sample from site 6.

of Middle Proterozoic paleopoles (Irving and McGlynn 1981), far distant from the B paleopole. Evidently the B magnetization is older than the latest Middle Proterozoic (not younger than about 1450 Ma), and younger than the "post-Hudsonian overprints" (1750–1600 Ma). These age limits (1600–1450 Ma) are consistent with the Rb–Sr age cited above. An early Middle Proterozoic or late Middle Proterozoic age is indicated for the B magnetization.

C magnetization

The C magnetization is found mainly in sandstones containing dark purple hematitic bands and reddish leisegang rings in the Midwest Lake – Rabbit Lake area. It also occurs sporadically in dark purple banded sandstones from Key Lake. Petrographic evidence indicates that this is the youngest hematite present. A Rb–Sr model age of 970 Ma has been obtained from diagenetic illite in the Rabbit Lake area displaying a similar paragenesis (Kotzer and Kyser 1990*b*).

The C paleopole (Fig. 9) is close to those for the Jacobsville (J2) and Chequamegon sandstones (CH). These magnetizations are borne by authigenic hematite in rocks of late Keweenawan age. Thus an age of early Late Proterozoic, comparable to the Rb–Sr age cited above, is probable.

Specific comparisons

As already noted, Athabascan paleopoles cannot yet be satisfactorily matched with well-dated paleopoles for Laurentia. However, there are two results of potential interest for dating A and B magnetizations. Both are from dyke swarms on the eastern shore of Great Bear Lake near Port Radium. Both swarms are unmetamorphosed and postdate the Great Bear Batholith (1850 Ma) and the main Early Proterozoic deformation of the Wopmay Orogen.

The first is a preliminary result from the Cleaver dyke swarm (one dyke only), which is not directly dated. It contains a single high unblocking temperature magnetization and has normal polarity (Fig. 10 in Irving *et al.* 1972). Its paleopole falls on the eastern limit of the group of late Early Proterozoic paleopoles (CL; Fig. 9). Potentially, an accurate age and full paleomagnetic survey of this dyke swarm could assist in dating the A magnetization.

The second result is from the Western Channel diabase intrusions that crosscut the Cleaver dykes (Hildebrand 1980). Both dykes and sills are represented, and the dykes yield a paleopole close to that from the B magnetization and with the same reversed polarity (WCD; Fig. 9). The swarm contains a dyke-like body that extends about 40 kms in a northeast–southwest direction and passes close to Port Radium. It has a paleopole (WCP; Fig. 9) that is not significantly different from the B paleopole. The diabase has yielded a K–Ar minimum age of 1400 ± 75 Ma (Wanless *et al.* 1970). Its age is therefore poorly defined. An accurate age could assist in dating the B magnetization more accurately.

The association of the B magnetization with uranium deposition and its similarity to that of the diabase intrusions at Port Radium raises the possibility that the fluid events associated with uranium mineralization at both places may be coeval. The possible association in time of both A and B magnetizations with intrusion of dyke swarms indicates that the tectonic events responsible for fluid flow may have been produced in part during intervals of crustal extension.

Summary and discussion

Oxygen-isotope and paleomagnetic measurements of hematite and clay minerals in the Athabasca Basin indicate that three magnetically distinct generations of hematite (A, B, and C) have formed during three separate fluid events. The fluids

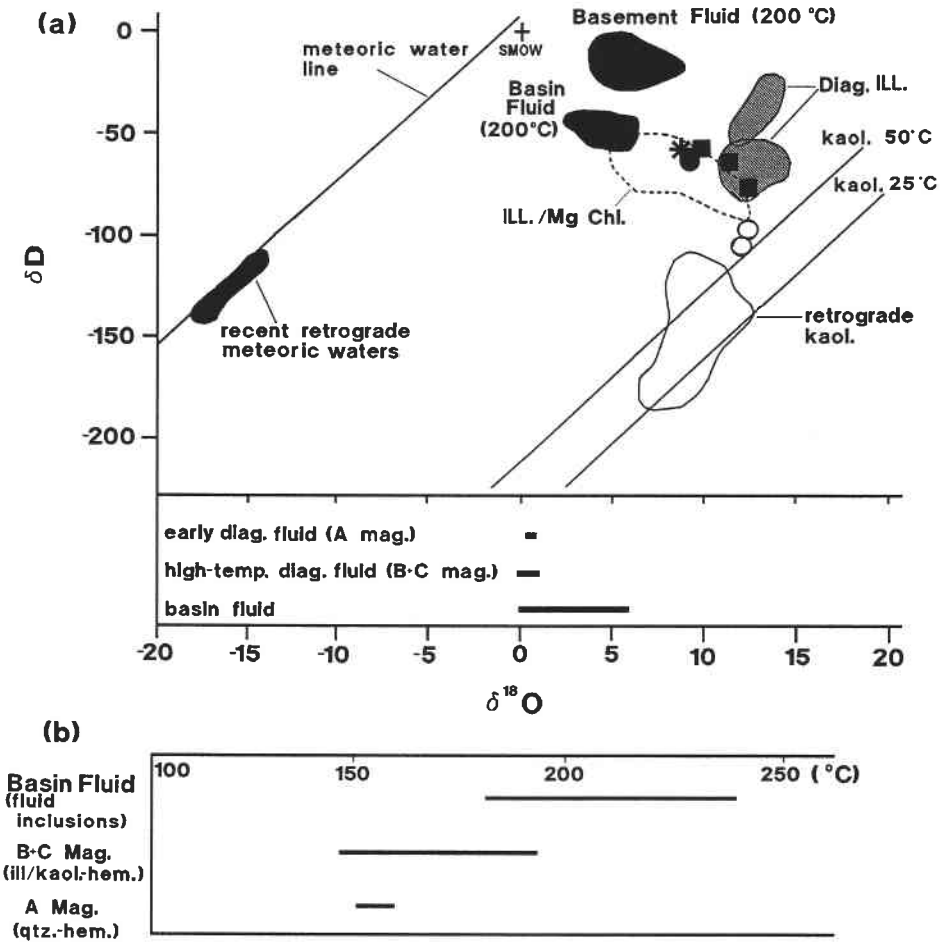


FIG. 8. (*a*) $\delta^{18}O$ and δD values of diagenetic clay minerals in samples collected for paleomagnetic analysis from Key Lake and Midwest Lake – Rabbit Lake area. ■, illite–kaolinite at Key Lake; ●, magnesium chlorite at Key Lake; ○, illite–kaolinite at Midwest Lake – Rabbit Lake area; *, illite at Eagle Point having a 970 Ma Rb–Sr model age. Shaded fields are from Wilson and Kyser (1987) and indicate ranges of isotopic values for (*i*) diagenetic illite and illite–kaolinite (Diag. ILL.) and fluid in equilibrium with them at 200°C (Basin Fluid) in Athabasca Basin, (*ii*) illite – magnesium chlorite mixtures in hydrothermally altered paleoregolith (ILL./Mg Chl.) and fluid in equilibrium with them at 200°C (Basement Fluid), and (*iii*) retrograde kaolinite in reactivated faults and late fractures hosting remobilized uranium (retrograde kaol.) and modern-day meteoric fluids in equilibrium with the kaolinite at temperatures of <50°C (recent retrograde meteoric waters). Diagram also shows ranges in $\delta^{18}O$ values of fluids determined from clay minerals and silicates from previous isotope studies (basin fluid), and of fluids in equilibrium with the A- , B- , and C-type hematite using the $\delta^{18}O$ values of the hematite and magnetite–water fractionation curves. (*b*) Range in temperatures of fluids for diagenetic basin fluid from previous isotopic and microthermometric studies (Basin Fluid), and calculated temperatures for A, B, and C hematite using $\delta^{18}O$ values of coexisting hematite and clay minerals (B + C Mag.) and hematite coeval with early quartz (A Mag.).

had similar oxygen isotopic compositions and temperatures. These fluid events appear to have occurred (*i*) sometime during the latter part of the Early Proterozoic (1750–1600 Ma, A), (*ii*) during the early Middle Proterozoic (1600–1450 Ma, B), and (*iii*) in the late Precambrian at about 900 Ma (C). Although these estimates are subject to considerable uncertainties, the order in time of the fluid events is established, and it is clear that they were well spaced over a period on the order of 10^8 years. The magnetizations do not have the same polarity and, within each, reversals do not occur. This indicates that although the events span much time, they themselves may not have been of very long duration.

Paleomagnetism, diagenesis, and uranium mineralization

The oxygen isotopic data indicate that the specular hematite carrying the B and C magnetizations has formed from fluids with temperatures of approximately 140–200°C and $\delta^{18}O$ values of 0.1–1.2‰ (Table 3). This agrees well with basin fluid temperatures of 180–240°C and $\delta^{18}O$ values of 3 ± 3‰ as determined from quartz and illite associated with hematite carrying the B and C magnetism (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*). In contrast, hematite carrying the A magnetization formed from a relatively ^{18}O-depleted ($\delta^{18}O$ = 0.8–0.9‰) and somewhat cooler (about 150°C) fluid. It is probably among the earliest diagenetic fluids that interacted with the Athabasca sediments. Similar $\delta^{18}O$ values and temperatures for the early basin fluids have been calculated using the oxygen isotopic composition of early quartz overgrowths at temperatures determined from fluid-inclusion microthermometry (Kotzer and Kyser 1990*a*).

At the Key Lake mine site, hematite carrying the B magnetization is spatially associated with faults hosting the high-grade uranium ore. That the B magnetization marks the major uranium ore-forming event is evidenced by the agreement in the

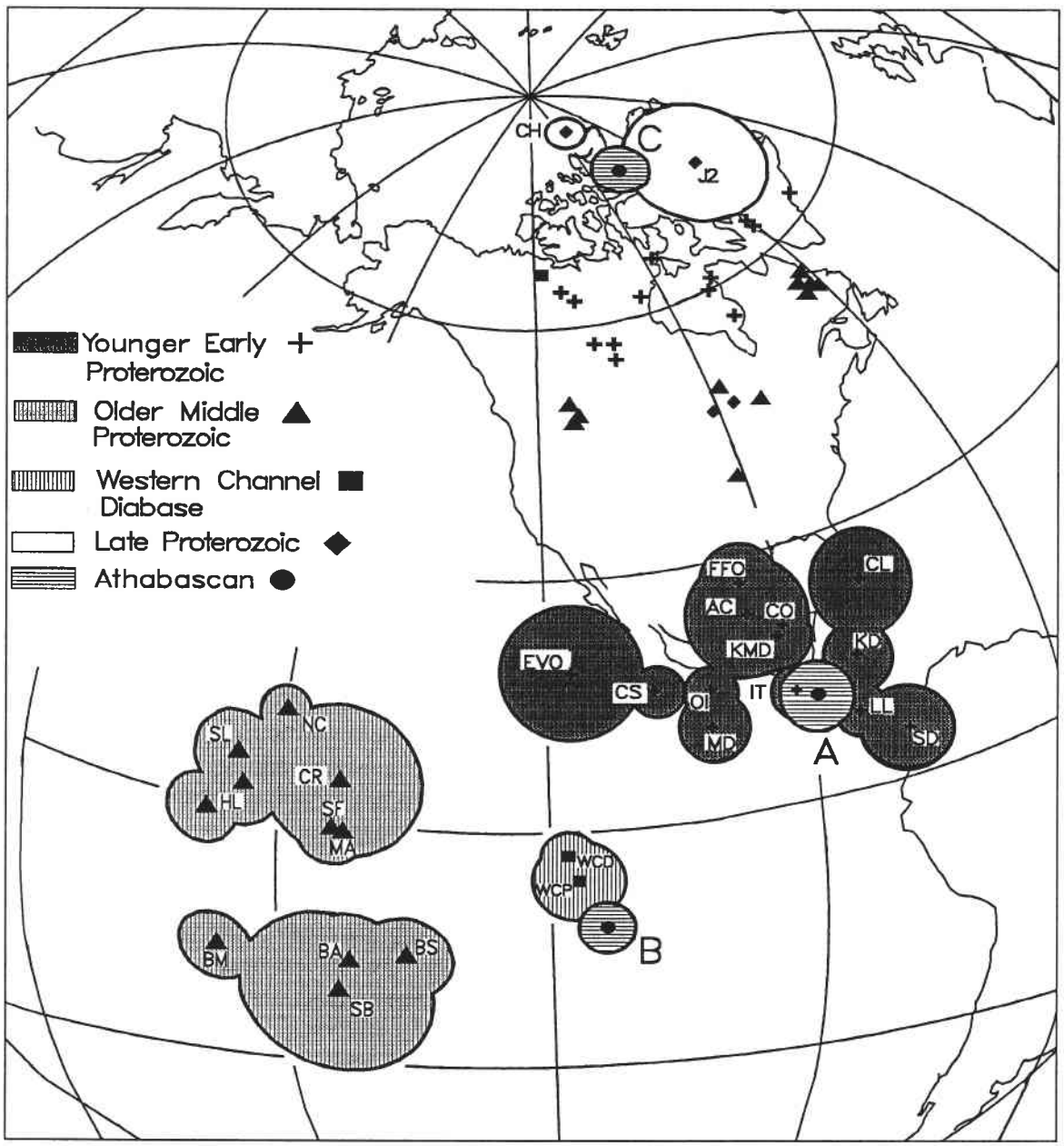


FIG. 9. Comparison of Athabasca paleopoles (A, B, C) with other Proterozoic paleopoles from Laurentia. The latter are arranged in three groups: (*i*) younger Early Proterozoic (1750–1600 Ma): AC, Angmagsslik Complex (Beckmann 1983); CL, Cleaver dykes (see below); CO, Coronation overprint (Evans *et al.* 1980); CS, Cape Smith basalt (Fujiwara and Schwarz 1975); EVO, Eskimo volcanics overprint (Schmidt 1980); FFO, Flin Flon overprint (Park 1975); IT, Itivdelq metamorphosed dykes and gneisses (Beckmann and Mitchell 1976; Morgan 1976); KD, Kangamuit metamorphosed dykes (Beckmann and Mitchell 1976); KMD, Kaminak metamorphosed dykes (Christie *et al.* 1975); LL, Lynn Lake gabbro (Dunsmore and Symons 1990); MD, Melville Peninsula – Daly Bay metamorphic rocks (Park 1973); OI, Ottawa Island komatiites (Buchan and Baragar 1985); SD, Slave dykes overprint (McGlynn and Irving 1975); (*ii*) older Middle Proterozoic (1480–1350 Ma): BA, Belt–Purcell Supergroup of Alberta (Evans *et al.* 1975); BM, Belt–Purcell Supergroup of Montana (Elston and Bressler 1980); BS, Belt–Purcell Supergroup, Spokane (Vitorello and Van der Voo 1977); CR, Croker Island Complex (Palmer 1969); HL, Harp Lake Complex (Irving *et al.* 1977); MA, Michickamau anorthosite (Emslie *et al.* 1976); MI, Mistastin pluton (Murthy *et al.* 1968; Emslie *et al.* 1976); NC, Nain Complex (Murthy 1978); SB, Sibley Group (Robertson 1973); SF, St. Francois igneous rocks (Hayes and Scharon 1966); SL, Seal Lake red beds (Roy and Fahrig 1973); WCD, Western Channel diabase dykes (Irving *et al.* 1972); WCP, the dyke through Port Radium part of WCD (see below); and (*iii*) Late Proterozoic overprints in Jacobsville Sandstone (J2) (Roy and Robertson 1978) and Chequamegon sandstone (CH) (McCabe and Van der Voo 1983). The Port Radium dyke (WCP) is situated at 66.3°N, 117.7°W; mean direction (*D*, *I*) is 359°, −32°, $n = 8$, $k = 117$, $\alpha_{95} = 5°$ obtained from 8 sites and paleopole 6°S, 117°W (calculated from data in Irving *et al.* 1972). The CL result has been obtained from 11 cores obtained from a prominent dyke of the Cleaver dyke swarm (66.1°N, 117.8°W) 1.3–2 km southeast of the mouth of Cameron Bay and 7.5 km east of Port Radium (Irving *et al.* 1972; site A144 and unpublished data). The mean direction (*D*, *I*) is 132°, 60°, $n = 11$, $k = 96$, $\alpha_{95} = 5°$ and paleopole 23°N, 80°W.

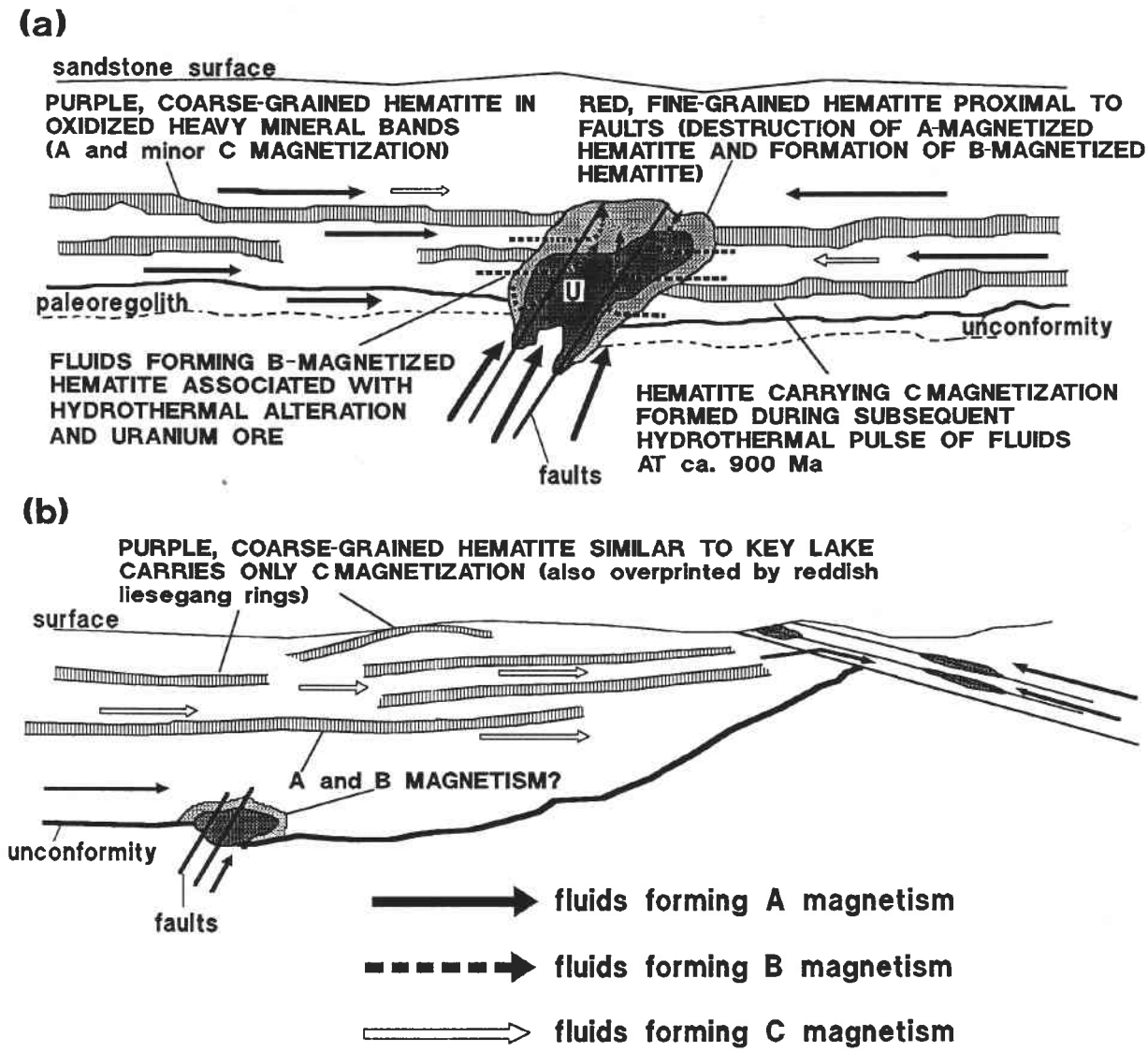


FIG. 10. Models for fluid flow and diagenetic hematite development based on paleomagnetic, geochemical and isotopic studies. (*a*) Key Lake (A, B, and C magnetism). A magnetism strongly developed in sandstones and paleoregolith by early diagenetic fluids at 1600−1750 Ma. Near major fault zones hematite carries only B magnetism, which corresponds with a major basin-basement fluid event associated with uranium mineralization at 1400−1500 Ma. C magnetism found sporadically in hematitic sandstones is associated with subsequent fluid pulse at approximately 900 Ma. (*b*) Midwest Lake − Rabbit Lake area (C magnetism only). C magnetism strongly developed in hematitic sandstones due to fluid pulse at approximately 900 Ma, which coincides with 970 Ma Rb−Sr model ages for illite from this area. No A or B magnetism was detected at these sites that may be due to (*i*) destruction of early formed hematite, (*ii*) local permeability restricting fluid flow of A and B events, or (*iii*) differences in fluid flow at various stratigraphic levels.

radiometric ages of uraninite and illite and the age of B magnetization determined paleomagnetically.

Fluid-flow patterns

In the Athabasca Basin, the regional and episodic nature of high-temperature fluid flow (basement and basin fluids) can be attributed to post-Hudsonian tectonic readjustment of the basin resulting in periodic reactivation of fault zones and selectively enhanced permeability. Stable isotope evidence indicates these fluids either originated from basement metasedimentary rocks or were evolved basin fluids within the sandstones. The latter appear to have predominated (Wilson and Kyser 1987; Kotzer and Kyser 1990*a*). That there was a long time interval between the formation of the hematite carrying the B and C magnetizations (ca. 600 Ma), both of which formed from fluids having similar $\delta^{18}O$ values, indicates that recharging of the basal sandstone aquifers by isotopically distinct fluids (such as meteoric waters) did not occur during this interval. It also indicates that fluid-flow events producing high-temperature diagenetic minerals (including uranium) were episodic and that permeability, although it must have been restricted, was not entirely destroyed by earlier generations of diagenetic minerals.

Similarities in the $\delta^{18}O$ values of hematite carrying the B

and C magnetism and in $\delta^{18}O$ values of coeval clay minerals indicate that the fluids from which they formed were chemically and isotopically similar. These hot, saline basin fluids were very widespread, as documented by fluid-inclusion and stable isotope studies on paragenetically equivalent silicate minerals (Pagel *et al.* 1980; Wallis *et al.* 1983; Wilson and Kyser 1987; Kotzer and Kyser 1990*a*, 1990*b*).

The spatial extent of the magnetizations delineating these fluid events can be estimated by comparing the results of the present study with previous paleomagnetic work on drill core taken from across the basin (Fig. 1) (Fahrig *et al.* 1978). The drill core was unoriented, so inclination but not declination was determined. The magnetic inclinations mainly fall into two groups with significantly different means of $62 \pm 12°$ and $-41 \pm 16°$. These are not significantly different from the $62 \pm 5°$ and $-37 \pm 7°$ obtained in this study for the A and B magnetizations (Table 2). Also a few specimens had steep positive inclinations of about 85° which could correspond to the C magnetization of this study. Thus, it appears that A, B, and possibly also C magnetizations are present across the entire basin. The drill-core data of Fahrig *et al.* (1978) also indicate that within a single drill hole, zones of A, B, and occasionally C magnetizations occur.

Because their results contained both positive and negative inclinations, Fahrig *et al.* (1978) interpreted them as evidence for geomagnetic reversals during initial deposition of the sediments. However, the hematite within the Athabasca Basin is diagenetic and possesses a CRM that was formed long after sedimentation. Moreover, the mean inclinations are not accurately opposed as would be expected from sequential reversals. Therefore, the positive and negative inclinations are not evidence of sequential reversals recorded during sedimentation but instead map the migration paths of temporally distinct diagenetic fluids.

A magnetizations occur in sandstone from the base of the Manitou Falls Formation and paleoregolith near the unconformity at the Key Lake mine site but are absent near to the fault zones that contain only the B magnetizations (Fig. 10*a*). It is possible that close to the fault zones, the early formed hematite (A) was destroyed during the subsequent hematite forming event (B), possibly as a result of local pH-f_{O_2} changes in the fluids (Schwertmann and Murad 1983) or enhanced fluid flow.

The occurrence of only C magnetization in the Midwest Lake – Rabbit Lake area indicates there was a substantial fluid event at approximately 900 Ma (Fig. 10*b*) that occurred at a relatively higher stratigraphic level in the Manitou Falls Formation than that represented by samples from the Key Lake mine site. This fluid event must also have occurred, less pervasively, in the basal levels of the Manitou Falls Formation, because C magnetizations also occur at the Key Lake mine site. Evidently the stratigraphy and faulting of the sandstones controlled the lateral and vertical fluid flow to a large extent.

Age of Athabasca sandstone

As already noted, the paleomagnetically determined age (1750–1600 Ma) for the A magnetization, which is believed to be the product of earliest diagenesis, is comparable to the minimum U–Pb ages of 1700–1650 Ma for early diagenetic apatite (Cummings *et al.* 1987). Fluid inclusions associated with the formation of A hematite indicate temperatures of 120–150°C, suggesting that significant accumulations of sediments preceded early diagenesis. It is concluded that the age of the Athabasca sediments is likely to be greater than 1700 Ma and less than 1750–1780 Ma (Hudsonian metamorphism).

Ages of 1720 ± 6 to 1685 ± 4 Ma from diagenetic apatite in the Thelon Basin (Miller *et al.* 1989) and age limits of 1850 to 1660 ± 8 Ma for the Hornby Bay Group (Bowring and Van Schmus 1982; Bowring and Ross 1985) suggest similar ages of formation for these tectonically and sedimentologically similar basins.

Conclusions

Paleomagnetic study of successive generations of hematite in the sedimentary rocks of the Athabasca Basin, when integrated with the $\delta^{18}O$ values from hematite and coexisting clay and silicate minerals, can assist in the determination of fluid flow in the basin (Fig. 10). The existence of three magnetically and temporally distinct generations of hematite formed from basin fluids having similar $\delta^{18}O$ values and temperature ranges suggests that movements of these fluids through the basin fill have been episodic and regional in nature and have occurred over time spans on the order of 10^8 years. The occurrence of weak, incoherent magnetizations (D) in areas of the sandstones located near the reactivated faults suggests that the fault zones were highly permeable to the late low-temperature meteoric waters that altered the diagenetic hematite to goethite. Hence, the ultimate controls on the movements of the basin fluids appear to have been permeability barriers in the sediments themselves, with large, active fault zones playing a subordinate but locally important role.

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Annesley, I. R., and Madore, C. 1991. U–Pb zircon/sphene ages for quartzofeldspathic gneisses and Aphebian granite from the Wollaston Domain, Cree Lake (Hearne Province), Saskatchewan. Geological Association of Canada, Program with Abstracts, **16**: A3.

Armstrong, R. L., and Ramaekers, P. 1985. Sr isotopic study of Helikian sediment and diabase dikes in the Athabasca Basin, northern Saskatchewan. Canadian Journal of Earth Sciences, **22**: 399–407.

Becker, R. H., and Clayton, R. N. 1976. Oxygen isotope study of a Precambrian banded iron formation, Hammersly Range, Western Australia. Geochimica et Cosmochimica Acta, **40**: 1153–1165.

Beckmann, G. E. J. 1983. Paleomagnetism of some Precambrian rocks in southeast Greenland. Physics of the Earth and Planetary Interiors, **32**: 85–99.

Beckmann, G. E. J., and Mitchell, J. G. 1976. Paleomagnetic and geochronological work in central west Greenland. Earth and Planetary Science Letters, **30**: 269–280.

Bethke, C., and Marshak, S. 1990. Brine migrations across North America—the plate tectonics of groundwater. Annual Review of Earth and Planetary Science Letters, **18**: 287–350.

Bigeleisen, J., Pearlman, M., and Prosser, H. C. 1952. Conversion of hydrogenic materials to hydrogen for isotopic analysis. Analytical Chemistry, **24**: 1356–1357.

Bowring, S., and Ross, G. 1985. Geochronology of the Narakay Volcanic Complex: implications for the age of the Coppermine Homocline and MacKenzie igneous event. Canadian Journal of Earth Sciences, **22**: 774–781.

Bowring, S., and Van Schmus, W. 1982. Age and duration of igneous events, Wopmay Orogen, Northwest Territories, Canada. Geological Society of America, Abstracts with Programs, **14**: A449.

Buchan, K. L., and Baragar, W. R. A. 1985. Paleomagnetism of komatiitic basalts of the Ottawa Islands, N.W.T. Canadian Journal of Earth Sciences, **22**: 553–566.

Carl, C., Hoehndorf, F., Pechmann, E., *et al.* 1988. Geochronology of the Key Lake uranium deposit, Saskatchewan, Canada. Journal of Chemical Geology, **70**: 133.

Christie, K. W., Davidson, A., and Fahrig, W. F. 1975. The paleomagnetism of Kaminak dykes—no evidence of significant Hudsonian plate motion. Canadian Journal of Earth Sciences, **12**: 2048–2064.

Clayton, R., and Mayeda, T. K. 1963. The use of bromine pentafluoride in the extraction of oxygen from oxides and silicates for isotopic analysis. Geochimica et Cosmochimica Acta, **27**: 43–52.

Cumming, G., and Krstic, D. 1991. The age of unconformity-related uranium mineralization in the Athabasca Basin, northern Saskatchewan. Geological Association of Canada, Program with Abstracts, **16**: A27.

Cumming, G., Krstic, D., and Wilson, J. 1987. Age of the Athabasca Group, northern Alberta. Geological Association of Canada, Program with Abstracts, **12**: A35.

Dunoyer de Segonzac, G. 1970. The transformation of clay minerals during diagenesis and low-grade metamorphism: a review. Sedimentology, **15**: 281–346.

Dunsmore, D. J., and Symons, D. T. A. 1990. Paleomagnetism of the Lynn Lake gabbros in the Trans-Hudson Orogen and closure of the Superior and Slave cratons. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 215–228.

Elmore, R. D., Dunn, W., and Peck, C. Absolute dating of dolomitization by means of paleomagnetic techniques. Geology, **15**: 558–561.

Elston, D. P., and Bressler, S. L. 1980. Paleomagnetic poles and polarity zonation from the middle Proterozoic Belt Supergroup, Montana and Idaho. Journal of Geophysical Research, **85**: 339–355.

Emslie, R., Irving, E., and Park, R. 1976. Further paleomagnetic results from the Michikamau Intrusives, Labrador. Canadian Journal of Earth Sciences, **10**: 1485–1493.

Eslinger, E., and Savin, S. 1973. Mineralogy and oxygen isotope geochemistry of the hydrothermally altered rocks of the Ohaki-Broadlands, New Zealand geothermal area. American Journal of Science, **273**: 240–270.

Evans, M. E., Bingham, D. K., and McMurry, E. W. 1975. New paleomagnetic results from the Upper Belt-Purcell Supergroup of Alberta. Canadian Journal of Earth Sciences, **12**: 52–61.

Evans, M. E., Hoye, G. S., and Bingham, D. K. 1980. The paleomagnetism of the Great Slave Supergroup: the Akaitcho River Formation. Canadian Journal of Earth Sciences, **17**: 1389–1395.

Fahrig, W. F., Christie, K., and Freda, G. 1978. The paleolatitude and paleomagnetic age of the Athabasca Formation, northern Saskatchewan. *In* Current research, part C. Geological Survey of Canada, Paper 78-1C, pp. 1–6.

Fisher, R. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**: 295–305.

Frape, S. K., and Fritz, P. 1987. Geochemical trends for groundwaters from the Canadian Shield. *In* Saline water and gases in crystalline rocks. *Edited by* P. Fritz and S. K. Frape. Geological Association of Canada, Special Paper 33, pp. 19–38.

Fujiwara, Y., and Schwarz, E. J. 1975. Paleomagnetism of the Circum-Ungava Proterozoic Fold Belt (I): Cape Smith komatiitic basalts. Canadian Journal of Earth Sciences, **12**: 1785–1793.

Globerman, B. R., and Irving, E. 1988. Mid-Cretaceous paleomagnetic reference field for North America: restudy of 100 Ma intrusive rocks from Arkansas. Journal of Geophysical Research, **93**: 11 721 – 11 733.

Hayes, W. W., and Scharon, L. 1966. A paleomagnetic study of the Precambrian igneous rocks of southeast Missouri. Journal of Geophysical Research, **71**: 553–560.

Hildebrand, R. S. 1980. Echo Bay – MacAlpine Channel area. Geological Survey of Canada, Map 1546A.

Hildebrand, R. S. 1988. Implications of ash dispersal for tectonic models with an example from Wopmay Orogen. Geology, **16**: 1089–1091.

Hodych, J., Patzold, R., and Buchan, K. 1984. Paleomagnetic study of the transformation of oolitic goethite to hematite in iron ore. Canadian Journal of Earth Sciences, **21**: 127–130.

Hoehndorf, F., Lenz, H., von Pechmann, E., *et al.* 1985. Radiometric age determinations of samples of Key Lake uranium deposits. *In* Geology of uranium deposits. *Edited by* T. I. I. Sibbald and W. Petruk. Canadian Institute of Mining and Metallurgy, Special Volume 32, pp. 48–53.

Hoeve, J., and Quirt, D. 1984. Mineralization and host-rock alteration in relation to clay mineral diagenesis and evolution of the middle-Proterozoic Athabasca Basin, northern Saskatchewan, Canada. Saskatchewan Research Council, Technical Report 187.

Hoeve, J., and Sibbald, T. 1978. On the genesis of Rabbit Lake and other unconformity-type uranium deposits in northern Saskatchewan. Economic Geology, **73**: 1450–1473.

Hohndorf, A., Strnad, J. G., and Carl, C. 1989. Age determinations of basement units in the Key Lake Uranium Deposit area, Saskatchewan, Canada. *In* Uranium resources and geology of North America. *Edited by* International Atomic Energy Agency. Proceedings, Technical Committee Meeting, International Atomic Energy Agency, Saskatoon, Sask., pp. 381–411.

Irving, E., and McGlynn, J. C. 1976. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia. Philosophical Transactions of the Royal Society of London, Series A, **280:** 433–468.

Irving, E., and McGlynn, J. C. 1981. On the coherence, rotation and paleolatitude of Laurentia in the Proterozoic. *In* Developments in Precambrian geology 4: Precambrian plate tectonics. *Edited by* A. Kroner. Elsevier, Amsterdam, pp. 561–598.

Irving, E., and Strong, D. 1984. Paleomagnetism of Early Carboniferous Deer Lake Group, western Newfoundland: no evidence for mid-Carboniferous displacement of "Acadia." Earth and Planetary Science Letters, **69**: 379–390.

Irving, E., and Strong, D. 1985. Paleomagnetism of rocks from Burin Peninsula, Newfoundland: hypothesis of Late Paleozoic displacement of Acadia criticized. Journal of Geophysical Research, **90**: 1949–1962.

Irving, E., Donaldson, J., and Park, R. 1972. Paleomagnetism of the Western Channel Diabase and associated rocks, Northwest Territories. Canadian Journal of Earth Sciences, **9**: 960–971.

Irving, E., Emslie, R., and Park, J. 1977. Paleomagnetism of the Harp Lake Complex and associated rocks. Canadian Journal Earth Sciences, **14**: 1187–1201.

Kotzer, T., and Kyser, T. K. 1990*a*. The use of stable and radiogenic isotopes in the identification of fluids and processes associated with unconformity-type uranium deposits. *In* Modern exploration techniques. *Edited by* L. S. Beck and C. T. Harper. Saskatchewan Geological Survey, Special Publication 10, pp. 115–131.

Kotzer, T., and Kyser, T. K. 1990*b*. Fluid history of the Athabasca Basin and its relation to uranium deposits. *In* Summary of investigations 1990. Saskatchewan Energy and Mines, Saskatchewan

Geological Survey, Miscellaneous Report 90-4, pp. 153–157.

Kotzer, T., and Kyser, T. K. 1991. Retrograde alteration of clay minerals in uranium deposits: radiation-catalyzed or simply low-temperature exchange? Isotope Geoscience, **86**: 307–321.

Kyser, T. K., and Kerrich, R. 1990. Geochemistry of fluids in tectonically active crustal regions. *In* Fluids in tectonically active regimes of the continental crust. *Edited by* B. E. Nesbitt. Mineralogical Association of Canada, Short Course 18, pp. 133–215.

Kyser, T. K., and O'Neil, J. 1984. Hydrogen isotope systematics of submarine basalts. Geochimica et Cosmochimica Acta, **48**: 48–53.

Kyser, T. K., Kotzer, T., and Sibbald, T. I. I. 1990. Oxygen, U–Pb and Pb–Pb isotope systematics in uraninite from complex U–Au–PGE vein-type and unconformity-type U deposits in northern Saskatchewan. *In* Summary of investigations 1990. Saskatchewan Energy and Mines, Saskatchewan Geological Survey, Miscellaneous Report 90-4, pp. 64–69.

Land, L., and Dutton, S. 1978. Cementation of Pennsylvanian deltaic sandstone: isotopic data. Journal of Sedimentary Petrology, **48**: 1167–1176.

Lu, G., Marshak, S., and Kent, D. 1990. Characteristics of magnetic carriers responsible for late-Paleozoic magnetization in carbonate strata of the mid-continent, USA. Earth and Planetary Science Letters, **99**: 351–361.

Machado, N. 1990. Timing of collisional events in the Trans-Hudson Orogen: evidence from U–Pb geochronology for the New Quebec Orogen, the Thompson Belt, and the Reindeer Zone (Manitoba and Saskatchewan). *In* The Early Proterozoic Trans-Hudson Orogen. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 433–441.

McCabe, C., and Elmore, R. D. 1989. The occurrence and origin of Late Paleozoic remagnetization in the sedimentary rocks of North America. Reviews of Geophysics, **27**: 471–493.

McCabe, C., and Van der Voo, R. 1983. Paleomagnetic results from the Upper Keweenawan Chequamegon Sandstone: implications for red-bed diagenesis and late-Precambrian apparent polar wander of North America. Canadian Journal of Earth Sciences, **20**: 105–112.

McCabe, C., Van der Voo, R., and Ballard, M. M. 1984. Late Paleozoic remagnetization of the Trenton Limestone. Geophysical Research Letters, **11**: 979–982.

McGlynn, J. C., and Irving, E. 1975. Paleomagnetism of early Aphebian diabase dykes from the Slave structural province, Canada. Tectonophysics, **26**: 23–38.

Miller, A. R., Cumming, G. L., and Krstic, D. 1989. U–Pb, Pb–Pb, and K–Ar isotopic study and petrography of uraniferous phosphate-bearing rocks in the Thelon Formation, Dubawnt Group, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **26**: 867–880.

Miller, J. D., and Kent, D. V. 1988. Regional trends in the timing of Alleghenian remagnetization in the Appalachians. Geology, **16**: 588–591.

Morgan, G. E. 1976. Paleomagnetism of a slowly cooled plutonic terrain in western Greenland. Nature (London), **259**: 382–385.

Murthy, G. S. 1978. Paleomagnetic results from the Nain Anorthosite and their tectonic implications. Canadian Journal of Earth Sciences, **15**: 516–525.

Murthy, G. S., Fahrig, W. F., and Jones, D. L. 1968. The paleomagnetism of the Michikamau anorthosite intrusion, Labrador. Canadian Journal of Earth Sciences, **5**: 1139–1144.

Oliver, J. 1986. Fluids expelled tectonically from orogenic belts: their role in hydrocarbon migration and other geologic phenomena. Geology, **14**: 99–102.

Pagel, M., Poty, B., and Sheppard, S. M. F. 1980. Contributions to some Saskatchewan uranium deposits mainly from fluid inclusion and isotopic data. *In* Uranium in the Pine Creek Geosyncline. *Edited by* S. Ferguson and A. Goleby. International Atomic Energy Agency, Vienna, pp. 639–654.

Palmer, H. C. 1969. The paleomagnetism of the Croker Island Complex, Ontario, Canada. Canadian Journal of Earth Sciences, **6**: 213–218.

Pan, H., Symons, T., and Sangster, D. 1990. Paleomagnetism of the Mississippi Valley-type ores and host rocks in the northern Arkansas and Tri-State districts. Canadian Journal of Earth Sciences, **27**: 923–931.

Park, J. K. 1973. Paleomagnetism of metamorphic rocks of Daly Bay and Melville Peninsula, Churchill Structural Province. Canadian Journal of Earth Sciences, **10**: 1079–1088.

Park, J. K. 1975. Paleomagnetism of the Flin Flon – Snow Lake Greenstone Belt, Manitoba and Saskatchewan. Canadian Journal of Earth Sciences, **12**: 1273–1290.

Park, J. K., Irving, E., and Donaldson, J. 1973. Paleomagnetism of the Precambrian Dubawnt Group. Geological Society of America Bulletin, **84**: 859–870.

Ramaekers, P. 1978. The paleolatitude and paleomagnetic age of the Athabasca Formation, northern Saskatchewan—Discussion. *In* Current research, part C. Geological Survey Canada, Paper 79-1C, pp. 117–119.

Ramaekers, P. 1979. Stratigraphy of the Athabasca Basin. *In* Summary of investigations 1979. Saskatchewan Energy and Mines, Saskatchewan Geological Survey, Miscellaneous Report 79-10, pp. 154–160.

Ramaekers, P. 1981. Hudsonian and Helikian Basins of the Athabasca Region, northern Saskatchewan. *In* Proterozoic basins of Canada. *Edited by* F. Campbell. Geological Survey of Canada, Paper 81-10, pp. 219–233.

Robertson, W. A. 1973. Pole position of thermally cleaned Sibley Group sediments and its relevance to Proterozoic magnetic stratigraphy. Canadian Journal of Earth Sciences, **10**: 180–193.

Roy, J. L., and Fahrig, W. F. 1973. The paleomagnetism of the Seal and Croteau rocks from the Grenville front, Labrador: polar wandering and tectonic implications. Canadian Journal of Earth Sciences, **10**: 1279–1301.

Roy, J. L., and Park, J. K. 1974. The magnetization process of certain redbeds: vector analyses of chemical and thermal results. Canadian Journal of Earth Sciences, **11**: 437–471.

Roy, J. L., and Robertson, W. A. 1978. Paleomagnetism of the Jacobsville Formation and the Apparent Polar Wander Path for the interval 1100 to 670 M.Y. for North America. Journal of Geophysical Research, **83**: 1289–1304.

Ruhrmann, G., and Von Pechmann, E. 1989. Structural and hydrothermal modification of the Gaertner Uranium deposit, Key Lake, Saskatchewan, Canada. *In* Uranium resources and geology of North America. *Edited by* International Atomic Energy Agency. Proceedings, Technical Committee Meeting, International Atomic Energy Agency, Saskatoon, Sask., pp. 363–379.

Schmidt, P. 1980. Paleomagnetism of igneous rocks from the Belcher Islands, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **17**: 807–822.

Schwertmann, U., and Murad, E. 1983. Effect of pH on the formation of goethite and hematite from ferrihydrite. Clays and Clay Minerals, **31**: 277–284.

Tremblay, L. 1982. Geology of the uranium deposits related to the sub-Athabasca unconformity, Saskatchewan, Canada. Geological Survey of Canada, Paper 81-20.

Vitorello, I., and Van der Voo, R. 1977. Late Hadrynian and Helikian pole positions in the Spokane Formation, Montana. Canadian Journal of Earth Sciences, **14**: 67–73.

Wallis, R., Saracoglu, N., Brummer, J., and Golightly, J. 1983. Geology of the McClean uranium deposits. *In* Uranium exploration in the Athabasca Basin, Saskatchewan, Canada. *Edited by* E. M. Cameron. Geological Survey of Canada, Paper 82-11, pp. 71–110.

Wanless, R. K., Stevens, R. D., Lachance, G. R., and DeLabio, R. N. 1970. Age determinations and geological studies. Geological Survey Canada, Paper 69-2A.

Wilson, M., and Kyser, T. K. 1987. Stable isotope geochemistry of alteration associated with the Key Lake uranium deposit, Canada. Economic Geology, **82**: 1540–1557.

Wilson, M., Kyser, T. K., Meinhert, H., and Hoeve, J. 1987. Changes in the H-O-Ar isotopic composition of clays during retrograde alteration. Geochimica et Cosmochimica Acta, **51**: 869–878.

Witte, W., and Kent, D. 1990. The paleomagnetism of red beds and basalts of the Hettangian Extrusive Zone, Newark Basin, New Jersey. Journal of Geophysical Research, **95**: 17 533 – 17 545.

Zijderveld, J. D. A. 1967. AC demagnetization of rocks: analysis and results. *In* Methods in paleomagnetism. *Edited by* D. W. Collinson, K. M. Creer, and S. K. Runcorn. Elsevier, Amsterdam, pp. 254–286.